Job Matching without Price Discrimination*

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July 15, 2026

In many labor markets, firms do not price discriminate among their workers. We study how a labor market with uniform salaries matches workers to jobs. To do so, we construct a job matching model in which each firm views workers as interchangeable and must pay all its workers the same salary. While an efficient stable outcome always exists, inefficient outcomes can be stable as well. Workers' preferred stable outcome is efficient. Firms prefer other (potentially inefficient) stable outcomes in which they pay lower salaries

Keywords: job matching, market power, uniform pricing, monopsony.

JEL codes: D43, D47; J3.

*For their excellent feedback we thank Shani Cohen, Vincent Crawford, Jerry Green, Scott Kominers, Robin Lee, Shengwu Li, Alessandro Lizzeri, Pietro Ortoleva, Michael Ostrovsky, Wolfgang Pesendorfer, Jesse Shapiro, Leeat Yariv, the attendees of the Harvard and Princeton theory workshops, and several anonymous referees. Silbert gratefully acknowledges financial support from the William S. Dietrich II Economic Theory Center. An earlier draft of this paper was circulated as 'Stable Matching in Monopsonistic Labor Markets'.

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1 Introduction

Most workers do not bargain before accepting a job offer (Hall & Krueger, 2012). Rather, their job pays a uniform salary that all workers in that job receive. This paper studies how a labor market with uniform salaries matches workers to jobs.

To do so, we connect the job matching literature to the labor monopsony literature. Like the job matching literature, we study stable outcomes: a matching of workers to firms, along with a salary schedule, from which no worker-firm coalition can profitably deviate. Canonical job matching models assume that each worker's salary can be set independently of her colleagues' salaries (Kelso & Crawford, 1982; Hatfield & Milgrom, 2005). In other words, they allow firms to perfectly price discriminate between workers. We instead require that each firm pay all its workers the same salary, as is assumed by canonical labor monopsony models (Robinson, 1933; Boal & Ransom, 1997; Manning, 2011). It is well-established in that literature that the uniform salary requirement can distort the allocation of workers across the labor market.

Our first contribution is to use a stable matching lens to provide a novel characterization of this distortion. We show that imposing the uniform salary restriction can generate inefficient stable outcomes, that these distortions never misallocate workers *conditional on firm sizes*, and that these distortions require two-sided heterogeneity.

Our second contribution is to show that nonetheless an efficient stable outcome always exists. Moreover, relative to any other stable outcome, this efficient stable outcome pays higher, is better for all workers, and is worse for all firms.

Our final contribution is to demonstrate when and how centralized matching can ameliorate the inefficiencies generated by uniform salaries. There exists a strategyproof mechanism that can elicit workers' preferences and implement an efficient stable outcome, provided that firms' production technologies are public information. There exists no such mechanism that can elicit firm production technologies.

Model. Our model comprises discrete sets of *workers* and *firms*. Each worker can be employed by at most one firm, while each firm can hire any number of workers. Workers have quasi-linear preferences over their salary and their firm. Firms' production technologies view workers as interchangeable and have decreasing differences[1]

An *outcome* comprises a *matching* of workers to firms and a *salary schedule* associating each firm with a salary. An outcome is *individually rational* if all agents' payoffs are non-negative. An individually rational outcome is *stable* if no firm and set of workers can deviate from the outcome and, at some salary, be better off. Note that we modify the usual definition of stability

[1]We show in Appendix D that when workers are not interchangeable in production, but salaries are still uniform within each firm, a stable outcome may not exist.

to respect the uniform salary restriction: all deviating workers must receive the same salary. A matching is *efficient* if it maximizes the sum of firm and worker payoffs. A matching is *hedonic efficient* if it maximizes the sum of firm and worker payoffs given firm sizes. Given that workers are interchangeable in production, hedonic efficiency asks whether the matching maximizes the total value of workers' non-wage amenities, holding production fixed.

The following example illustrates many results, which we will see hold more generally.

Example 1. *Consider a labor market comprising a single firm and two workers: Jesse and Wilbur. The firm has a revenue production function $y(N) = 6N$, where N is the number of employed workers. Letting s denote the firm's salary, worker Jesse has utility function*

$$u_{Jesse} = \begin{cases} s & \text{if Jesse is employed} \\ 0 & \text{otherwise,} \end{cases}$$

while worker Wilbur has utility function

$$u_{Wilbur} = \begin{cases} s-4 & \text{if Wilbur is employed} \\ 0 & \text{otherwise.} \end{cases}$$

The example's stable outcomes are presented in Figure 1. First consider the matching in which both workers are employed. This matching is efficient: each employed worker generates 6 units of output but at most 4 units of worker disutility. This efficient matching will also be stable, provided it is coupled with a salary $s \in [4,6]$. Such a salary ensures that the firm makes non-negative profits and that both workers are at least as well off as they would be unemployed. Moreover, no other outcome would both yield the firm greater profits and yield greater utility to the still-employed workers.

There is, however, an *inefficient* matching that can be stable as well. Consider the matching in which only *Jesse* is employed, along with a salary $s \in [0,2)$. Such an outcome would yield the firm a profit of $6-s > 4$. Enticing *Wilbur* into employment would require that he be paid a salary of at least 4. But because *Jesse* would also have to be paid the new salary, the firm would then make profit of only $2(6-s) \leq 4$. The outcomes without *Wilbur* are thus stable.

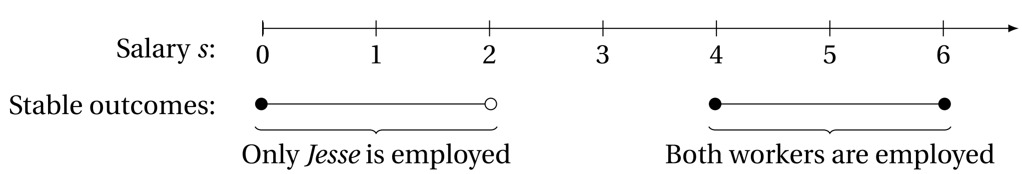


Figure 1: Stable outcomes in the Simple Example

We have shown that there are two distinct matchings associated with stable outcomes — even though only one of the matchings is efficient. Our model thus behaves very differently from the Kelso and Crawford model, which always predicts efficient matchings. Note that the inefficient matching can be in a stable outcome only because of the uniform salary restriction. If the firm could pay different salaries to each worker, the firm could continue to pay *Jesse* his current salary while offering *Wilbur* a salary of 4, profitably employing both workers. Thus, it is exactly the restriction on salaries that creates the distortion: the firm prefers employing inefficiently few workers over efficiently employing both workers at a higher salary.

Yet, the uniform salary restriction does not *prevent* the efficient matching from being stable. Although the firm prefers the outcomes in which it only employs *Jesse* at a lower salary, a firm which found itself paying a higher salary would be incentivized to employ both workers.

Example 1 exhibits other results that hold more generally. Although the outcomes in which only *Jesse* is employed are inefficient, they *are* hedonic efficient: given that only one worker is employed, it is more efficient for that worker to be *Jesse*. In addition, when $s = 6$ — a higher salary than in any other stable outcome — the outcome is efficient, is better for all workers than any other stable outcome, and yields lower firm profit than any other stable outcome.

Efficiency and stability. Our first major result is that there will *always* exist an efficient stable outcome. As we prove that result, we also characterize efficiency and stability more generally.

Stability is equivalent to three properties: No Envy, No Firing, and No Poaching. An outcome has *No Envy* if, given the salaries, each worker prefers her firm to any other firm. An outcome has *No Firing* if, given its salary, no firm would be better off matched to fewer workers. An outcome has *No Poaching* if no firm can increase its salary and make at least as much profit by attracting more workers. An outcome is stable if and only if it has No Envy, No Firing, and No Poaching.

With a discrete production function, a firm's marginal product can be defined either as the increase in output from hiring an additional worker or as the decrease in output from firing a single worker. An outcome has *Marginal Product Salaries* if every firm's salary lies within those two bounds. An outcome with Marginal Product Salaries will have No Firing and No Poaching. An outcome with both Marginal Product Salaries and No Envy will thus be stable.

To prove that an efficient stable outcome always exists, we start from an efficient matching and construct an outcome with both Marginal Product Salaries and No Envy. To do so, we set salaries maximally, subject to the constraint implied by No Envy and the constraint that no firm's salary is greater than its marginal product (from firing a worker). We use the efficiency of the matching to prove that each of these maximal salaries is also no *less* than the firm's marginal product (from hiring a worker). The constructed outcome thus has both Marginal Product Salaries and No Envy, and so will be stable.

Indeed, we show that *any* stable outcome with Marginal Product Salaries will be efficient.

However, not every stable outcome has Marginal Product Salaries, which is why not every stable outcome is efficient. For example, in the inefficient stable outcomes of Example 1, the firm did not want to poach *Wilbur*, even though its salary was less than its marginal product.

These relationships between stability and efficiency are summarized in Figure 2: a matching is efficient if and only if there exists an associated outcome with both No Envy and Marginal Product Salaries. Such an outcome will also have No Firing and No Poaching, and so is stable.

Although stable outcomes need not be efficient, the last arrow of Figure 2 shows that they are nonetheless hedonic efficient. Given workers are interchangeable in production, hedonic efficiency is only a function of workers' non-wage preferences, the sum total of which are maximized by the No Envy condition.

As Figure 2 makes clear, inefficiencies arise because of the gap between the No Firing and No Poaching conditions and the Marginal Product Salaries condition — a gap which does not appear in existing job matching models. In particular, No Poaching requires that a firm not find it profitable to poach an additional worker if it meant increasing its inframarginal workers' salaries, while Marginal Product Salaries requires that a firm not find it profitable to poach an additional worker based only on that worker's marginal productivity. Accordingly, we show that, in any inefficient stable matching, there will exist a worker that some firm would have poached, had the firm been able to price discriminate.

To further elucidate why inefficiencies arise, we consider two conditions under which no such worker can exist, and so every stable outcome is efficient. The first such condition is *Common Value Amenities*, which requires that workers have homogeneous preferences over the firms. Under Common Value Amenities, each worker is equally expensive, so firms wanting to

1: The matching μ is *efficient*

$\Updownarrow$

2: $\exists$ salary schedule s such that (μ, s) has *No Envy* and *Marginal Product Salaries*

$\Downarrow$

3: $\exists$ salary schedule s such that (μ, s) has *No Envy*, *No Firing* and *No Poaching*

$\Updownarrow$

4: $\exists$ salary schedule s such that (μ, s) is *stable*

$\Downarrow$

5: The matching μ is *hedonic efficient*

Figure 2: Characterization of Stability and Efficiency

expand would have no need to price discriminate.[2] The second such condition is that each firm has a *duplicate*: another firm with both the same production function and which provides the same amenities. Given that stable outcomes have No Envy, two duplicate firms must both pay the same salary. This means that each could poach the other's workers by paying an infinitesimally higher salary; again, this condition ensures that a firm can expand without needing to price discriminate.

Worker and firm welfare across stable outcomes. Recall the efficient stable outcome we constructed to prove that such an outcome always exists. Our second main result is that this efficient outcome has greater salaries than any other stable outcome, is preferred by all workers over any other stable outcome, and is *less* preferred by all firms than any other stable outcome. In this sense, workers' preferences are aligned with efficiency whereas firms' need not be.

Indeed, we show that all workers prefer one stable outcome over another if and only if the former has higher salaries. We also show that, if all workers prefer one stable outcome over another, then all firms must prefer the latter. However, in contrast to the typical dual lattice results in the two-sided matching literature (Knuth, 1976; Shapley & Shubik, 1971; Hatfield & Milgrom, 2005; Blair, 1988), the converse does not hold: there may be a stable outcome that all firms and some of the workers prefer over another stable outcome.

Among stable outcomes, there is no trade-off between worker welfare and market efficiency: the worker-optimal stable outcome is efficient. It is thus a promising target for a centralized matching mechanism.

Mechanism design. Our final results show when an efficient stable outcome can be implemented through a strategyproof mechanism. Our model contains two potential sources of private information: workers' preferences for employment at each firm, and the production functions with which firms produce their outputs. When firms' production functions are private information, no strategyproof mechanism can always implement an efficient stable outcome. Firms can claim that they are less productive than they actually are. This deceit results in them paying lower salaries, and thus can be profitable even if it means that they are matched to inefficiently few workers. However, when firms' production functions are public information, a strategyproof mechanism can elicit workers' preferences and implement the worker-optimal efficient stable outcome. This result demonstrates when centralized matching can ameliorate the distortions arising from uniform salaries: only when firms' production functions are known.

Literature review. This project unifies the job matching and the labor monopsony literatures. The intellectual antecedent of job matching models is the Gale and Shapley (1962) college admissions model, in which students' preferences over colleges are combined with colleges'

[2]The parallel to the labor monopsony literature is that, under Common Value Amenities, labor supply curves are not upward-sloping.

preferences over students to construct a stable matching: that is, a matching from which no set of students and colleges can profitably defect. Job matching models extend the college admissions model by pairing each matching with transfers from one side of the market to the other (Shapley & Shubik, 1971; Crawford & Knoer, 1981; Kelso & Crawford, 1982; Fleiner, 2003; Hatfield & Milgrom, 2005).

The canonical job matching model is that of Kelso and Crawford (1982). Kelso and Crawford show that, if firms treat workers as 'gross substitutes' and transfers are unrestricted, then a stable outcome always exists. When workers' utilities are quasi-linear in salaries, every stable outcome of the Kelso and Crawford model is efficient. Kelso and Crawford assume that each worker's salary can be set independently of the salaries paid to that worker's colleagues. Thus, a blocking coalition consisting of one worker and one firm will not affect the transfers paid to other workers whom that firm employs. Such coalitions block any inefficient outcome.

Although some existing job matching models constrain the allocation of workers to firms (Kojima, Sun, & Yu, 2020), to the best of our knowledge ours is the first to constrain salaries.[3] In contemporaneous work, Gentile Passaro, Kojima, and Pakzad-Hurson (2026) present a model in which firms must offer uniform salaries if their workers are from different groups, but can price discriminate otherwise. Firms in the Gentile Passaro et al. model generically become perfectly segregated, so retain the ability to price discriminate.

Given that we constrain salaries, stable outcomes in existing job matching models need not be stable outcomes in our model. Moreover, blocking coalitions in existing models need not be blocking coalitions in our model. Our stable outcomes are neither a subset nor a superset of stable outcomes in existing models, and so our results do not follow from existing results.

The labor monopsony literature descends from Robinson's (1933) study of imperfect competition. Modern monopsony models adopt functional form restrictions more frequently than the job matching literature. For example, some models postulate a representative worker with CES labor disutility (Berger, Herkenhoff, & Mongey, 2019). Others postulate a continuum of workers with Gumbel-distributed firm amenities (Card, Cardoso, Heining, & Kline, 2018; Lamadon, Mogstad, & Setzler, 2019; Kroft, Luo, Mogstad, & Setzler, 2020). Firms interact in Bertrand or Cournot competition, with or without search frictions, and a firm either pays an identical salary to all its workers or discriminates solely on the basis of productivity. A recurring theme is that firms' strategic behavior distorts the labor market: unemployment is too high, productive firms

[3] Hatfield, Kominers, Nichifor, Ostrovsky, and Westkamp (2013) generalize job-matching models by allowing for more general network structures. Their results extend to 'anonymous' prices: a buyer of perfectly-substitutable goods can be assumed to face the same price for each. As is clear from contrasting our results with theirs (e.g. with respect to efficiency), this generalization only holds for 'mutually incompatible' goods – e.g., when a 'firm' can only hire a single 'worker'.

are too small and unproductive firms are too large (Boal & Ransom, 1997; Manning, 2011; Berger et al., 2019; Lamadon et al., 2019).

These distortions are not found in Kelso and Crawford's model. Given the different modelling assumptions made by the job matching and monopsony literatures, it is not prima facie obvious why that is: Is it because of the functional forms they impose? Because of the solution concepts they employ? Or is it only because the job matching literature allows price discrimination? By unifying the two literatures, this paper demonstrates how uniform salaries alone can distort the allocation of workers – the first of our two main contributions.[4]

By studying market power in job matching, we follow Bulow and Levin (2006); Kojima (2007), and Azevedo (2014). Bulow and Levin consider only one-to-one matching. Their model thus lacks the 'monopsonistic' mechanism which stabilizes inefficient matchings in our model. Kojima (2007) extends Bulow and Levin's model by allowing each hospital to be matched to many doctors, but limits his comparisons to the firm-optimal competitive equilibrium. Our results in Section 5 C suggest that this perspective is limiting. Azevedo (2014) either treats salaries as exogenous or lets salaries vary between the workers employed by a given firm. These assumptions also foreclose the distortion on which we focus.

Roadmap. Section 2 presents our model and provides a succinct characterization of stability. Section 3 studies the relationship between stability and efficiency. Section 4 studies exactly how inefficiencies arise in a model with uniform salaries. Section 5 studies worker and firm welfare across the set of stable outcomes. Section 6 asks when an efficient outcome can be implemented using a strategyproof mechanism. Section 7 concludes. Appendix A contains proofs of main results. An online appendix contains the remaining proofs and some additional results.

2 A Model of a Labor Market

A labor market $(\mathbf{F}, \mathbf{W})$ comprises a finite set of firms $\mathbf{F}$ and a finite set of workers $\mathbf{W}$. Each worker can be employed by at most one firm while each firm can hire any number of workers.

Each firm F is endowed with a production technology, which we represent with a non-decreasing function $y_F : \mathbb{N} \to \mathbb{R}^+$. Note that production depends only on the number of workers employed and not on their identity. We normalize each $y_F(0) = 0$. Each firm pays the same salary to all its workers: there is no salary discrimination within any firm's workforce. Firms face

[4]Whether equally-productive workers actually are paid equally is an active research topic in empirical labor economics, with conflicting results from both surveys (Hall & Krueger, 2012; Caldwell, Haegele, & Heining, 2026) and quasi-experiments (Townsend & Allan, 2025; Carvalho, Galindo da Fonseca, & Santarrosa, 2023). We expect future empirical work will clarify when the uniform salaries assumption is justified and so clarify when the results in this paper are relevant.

a competitive product market in which their good has price normalized to one. Thus, if firm F employs N workers at salary s, its profit is

$$\pi_F(N,s) = y_F(N) - sN.$$

Each worker $w \in \mathbf{W}$ has quasi-linear preferences

$$u_w(F,s) = \alpha_w(F) + s,$$

where $F \in \mathbf{F} \cup \{\emptyset\}$ is the firm at which she is employed, s is the salary she is paid, and $\alpha_w(F)$ is the amenity that she receives from working at firm F. The amenity $\alpha_w(F)$ may be positive, negative, or zero, and can vary idiosyncratically across workers. It encompasses any fixed benefit or cost the worker incurs from working at a given firm. Being employed at the empty set denotes unemployment, and we normalize $\alpha_w(\emptyset) = 0$.

2.1 Matchings and outcomes

A **matching** is a function $\mu : \mathbf{F} \cup \mathbf{W} \to \mathscr{P}(\mathbf{F} \cup \mathbf{W})$ such that:

- For all workers $w \in \mathbf{W}$: $\left|\mu(w)\right| \leq 1$ and $\mu(w) \subseteq \mathbf{F}$.
- For all firms $F \in \mathbf{F}$: $\mu(F) \subseteq \mathbf{W}$.
- For all workers $w \in \mathbf{W}$ and all firms $F \in \mathbf{F}$: $w \in \mu(F)$ if and only if $\mu(w) = \{F\}$.

We use the matching to represent employment: a worker w is employed at firm F if and only if $\mu(w) = \{F\}$. Since workers are matched to at most one firm, we abuse notation and write $\mu(w) = F$ rather than $\mu(w) = \{F\}$.

An **outcome** $\left(\mu, s\right)$ comprises a matching μ and a salary schedule $s : \mathbf{F} \cup \{\emptyset\} \to \mathbb{R}^+$, associating each firm with a salary. We require all salaries to be non-negative, and we normalize $s(\emptyset) = 0$. To simplify our results we require that for any outcome $\left(\mu, s\right)$, for any firm F, if $\mu(F) = \emptyset$, then $s(F) = y_F(1)$. This requirement is without loss of generality because the salary paid by an unmatched firm does not affect its profit.

An outcome $\left(\mu, s\right)$ is **individually rational** if

- for all workers $w \in \mathbf{W}$: $u_w\left(\mu(w), s(\mu(w))\right) \geq 0$, and
- for all firms $F \in \mathbf{F}$: $\pi_F\left(\left|\mu(F)\right|, s(F)\right) \geq 0$.

This familiar condition requires that no worker receive less utility than she would from unemployment, and that no firm make negative profits.

A coalition (F, C, s^*), with $F \in \mathbf{F}$, $C \subseteq \mathbf{W}$, and $s^* \in \mathbb{R}^+$, **blocks** outcome $\left(\mu, s\right)$ if

- $\pi_F(|C|, s^*) \geq \pi_F\left(\left|\mu(F)\right|, s(F)\right)$, and
- for all workers $w \in C$: $u_w(F, s^*) \geq u_w\left(\mu(w), s(\mu(w))\right)$,

where the inequality is strict for the firm or one of the workers.[5] In words, a firm and some workers can block an outcome if there is a salary at which, if the firm employs only those workers, all parties are better off.

Our solution concept is stability[6] An outcome (μ, s) is **stable** if it is individually rational and not blocked by any coalition.

2.2 Production functions

We required above that firms' production functions are non-decreasing and normalized such that for all firms F: $y_F(0) = 0$. Following the matching literature (Kelso & Crawford, 1982; Hatfield & Milgrom, 2005), we additionally assume that firms treat workers as **gross substitutes**: For each firm $F \in \mathbf{F}$, salary $s \in \mathbb{R}^+$ and number of employees N:

$$N \in \underset{M \leq N}{\arg\max}\, \pi_F(M, s) \implies N - 1 \in \underset{M \leq N-1}{\arg\max}\, \pi_F(M, s).$$

In other words, if, at some salary, a firm prefers to hire N workers over any fewer number, it must also prefer to hire $N-1$ workers over any fewer number.

A function $f : \mathbb{N} \to \mathbb{R}$ has **decreasing differences** if $N > M$ implies that $f(N+1) - f(N) \leq f(M+1) - f(M)$. Theorem 6 in Kelso and Crawford shows that, when workers are interchangeable in production, gross substitutes is equivalent to the production functions y_F having decreasing differences.[7] Given that our notion of gross substitutes is slightly different from theirs, we provide our own proof. (Our proofs are in Appendix A.)

Lemma 1. *Every firm's production function has decreasing differences.*

With a discrete production function, a firm's marginal product has two possible definitions. Given some matching μ, we let $\Delta^+_\mu(F)$ denote the increase in firm F's output from employing

[5] It is without loss of generality to consider only blocking coalitions comprised of a single firm: any coalition containing more than one firm could be be broken into multiple coalitions, each with only one firm.

[6] Using stability as a solution concept does not imply that all stable outcomes are equally realistic as market equilibria. For example, one implication of stability is that firms cannot decrease their salaries without the consent of their current workers; the plausibility of some stable outcomes will thus depend on whether contracts or regulations enforce this feature. The labor literature often adopts non-cooperative solution concepts like Bertrand competition. In Appendix C, we show that Bertrand equilibria are generically stable.

[7] Typically, the gross substitutes condition is used to guarantee the existence of stable outcomes (Kelso & Crawford, 1982; Hatfield & Milgrom, 2005). This condition is often compared to the concavity of utility or production functions. In our setting, this connection becomes even more clear: when workers are interchangeable in production and are treated as gross substitutes, production functions exhibit diminishing marginal returns to labor. In this way, the gross substitutes assumption is the discrete analogue of assuming concave production.

one worker *more* than the firm is employing at μ, and we let $\Delta^-_\mu(F)$ denote the decrease in firm F's output from employing one worker *fewer* than the firm is employing at μ:

$$\Delta^+_\mu(F) \equiv y_F\left(\left|\mu(F)\right|+1\right) - y_F\left(\left|\mu(F)\right|\right);$$

$$\Delta^-_\mu(F) \equiv \begin{cases} y_F\left(\left|\mu(F)\right|\right) - y_F\left(\left|\mu(F)\right|-1\right) & \text{if } \mu(F) \neq \emptyset; \\ y_F(1) & \text{if } \mu(F) = \emptyset. \end{cases}$$

By Lemma 1, for any firm F and matching μ: $\Delta^+_\mu(F) \leq \Delta^-_\mu(F)$. We say that an outcome (μ, s) has **Marginal Product Salaries** if every firm's salary is between these two quantities:

$$\forall F \in \mathbf{F}: \; s(F) \in \left[\Delta^+_\mu(F), \, \Delta^-_\mu(F)\right].$$

Given that unemployed workers produce no output we say that the marginal product of unemployment is zero: $\forall \mu: \Delta^+_\mu(\emptyset) = \Delta^-_\mu(\emptyset) = 0$.

2.3 Definitions of efficiency

Given the quasi-linear payoffs, we define the **value** of a matching as the sum of worker amenities and firm outputs:

$$\text{value}\left(\mu\right) \equiv \sum_{F\in\mathbf{F}} y_F\left(\left|\mu(F)\right|\right) + \sum_{w\in\mathbf{W}} \alpha_w\left(\mu(w)\right).$$

A matching μ^* is **efficient** if it has maximal value:

$$\mu^* \in \operatorname*{arg\,max}_{\mu} \text{value}\left(\mu\right).$$

This notion of efficiency is sometimes referred to as utilitarian efficiency.

A matching μ^* is **hedonic efficient** if it maximizes the sum of amenities, given firm sizes:

$$\mu^* \in \operatorname*{arg\,max}_{\mu \text{ s.t.} \forall F: \left|\mu(F)\right| = \left|\mu^*(F)\right|} \sum_{w\in\mathbf{W}} \alpha_w\left(\mu(w)\right).$$

By holding firm sizes fixed, hedonic efficiency speaks only to inefficiencies caused by a mismatch of workers to firms, rather than inefficiencies in production. Note that hedonic efficiency is a strictly weaker requirement than efficiency: if μ^* is efficient, then μ^* is hedonic efficient.

An outcome (μ, s) is efficient if μ is efficient, and is hedonic efficient if μ is hedonic efficient.

2.4 Characterizing stable outcomes

We conclude this section by characterizing stability. An outcome (μ, s) has **No Envy** if for every worker $w \in \mathbf{W}$ and firm $F \in \mathbf{F} \cup \{\emptyset\}$:

$$u_w\left(\mu(w), s(\mu(w))\right) \geq u_w\left(F, s(F)\right).$$

The No Envy condition states that no worker would prefer to be matched to another firm, or to be unemployed, given the prevailing salaries.

An outcome (μ, s) has **No Firing** if for every firm $F \in \mathbf{F}$ such that $\left|\mu(F)\right| > 0$:

$$\pi_F\left(\left|\mu(F)\right|, s(F)\right) \geq \pi_F\left(\left|\mu(F)\right| - 1, s(F)\right).$$

The No Firing condition states that no firm would be better off being matched to one worker fewer while paying the same salary. Using the definition of the marginal product $\Delta_\mu^-(F)$ above, No Firing is equivalent to the requirement that for each matched firm, $s(F) \leq \Delta_\mu^-(F)$.

Consider an outcome in which firm F pays salary s_F and each other firm F' pays $s\left(F'\right)$, i.e., the corresponding element of the salary schedule s. The maximal labor-supply available for firm F is

$$L_F(s_F, s) \equiv \left|\left\{w : u_w(F, s_F) \geq \max_{G \in \mathbf{F} \cup \{\emptyset\}} u_w\left(G, s(G)\right)\right\}\right|.$$

An outcome (μ, s) has **No Poaching** if for every firm $F \in \mathbf{F}$, there exists no salary $s_F > s(F)$ and employment level L such that $\left|\mu(F)\right| < L \leq L_F(s_F, s)$, and that

$$\pi_F(L, s_F) \geq \pi_F\left(\left|\mu(F)\right|, s(F)\right).$$

The No Poaching condition states that no firm can increase its salary and, by attracting additional workers, make at least as much profit as it did previously.

Proposition 1. *An outcome is stable if and only if it has No Envy, No Firing, and No Poaching.*

If the No Envy condition fails, some 'envious' worker would prefer employment at some other firm, at that firm's existing salary. The envious worker could form a blocking coalition with that firm and all-but-one of that firm's existing workers, at the existing salary. The envious worker would be better off, while the firm and the existing workers would be no worse off. The fact that the firm is no worse off follows from our assumption that production depends only on the number of workers employed and not on their identity.

If the No Firing condition fails, some firm would be losing money on its marginal worker. The outcome would be blocked by a coalition comprising that firm and all-but-one of the firm's existing workers, at the existing salary. That coalition would leave the still employed workers no worse off and the firm would make strictly more profit.

If the No Poaching condition fails, some firm could form a blocking coalition with its existing workers and some new workers at a higher salary than it currently pays. The firm and the new workers would be no worse off, and the existing workers would be strictly better off.

The above summarizes why a stable outcome must have No Envy, No Firing, and No Poaching. To prove the converse, we first note that an outcome having No Envy must be individually rational for the workers. Given Lemma 1, the No Firing condition implies it is individual rational for the firms. We conclude the proof by showing that, if any blocking coalition exists, then the original outcome must not satisfy the No Envy, No Firing, or No Poaching conditions.

These three conditions encompass the ways in which a labor market outcome can be destabilized. Workers can quit or switch jobs (precluded by No Envy), firms can fire workers (precluded by No Firing), and firms can poach workers (precluded by No Poaching).

3 Efficiency and Stability

In this section, we study the relationship between efficiency and stability. We know from the Simple Example that stable outcomes need not be efficient. We show that at least one efficient outcome is always stable:

Theorem 1. *Let μ^* be an efficient matching. There exists a salary schedule s^* such that the outcome $\left(\mu^*, s^*\right)$ is stable.*

In other words, while the uniform salary restriction *can* distort the allocation of workers, it need not *necessarily* do so.

We also relate stability and efficiency more generally:

Proposition 2.

1. *If an outcome has No Envy and Marginal Product Salaries, then it is stable and efficient.*
2. *If an outcome is stable, then it is hedonic efficient.*

Marginal Product Salaries incentivize firms to maintain efficient employment levels. In addition, No Envy ensures that workers are allocated optimally given those employment levels. The first claim of Proposition 2 tells us that Marginal Product Salaries and No Envy together ensure that an outcome will be both stable and efficient. However the Simple Example showed us that Marginal Product Salaries are not *necessary* for an outcome to be stable, and thus the first claim of Proposition 2 does not imply that *every* stable outcome is efficient.

The second claim of Proposition 2 makes the weaker claim that every stable outcome is *hedonic* efficient. This result indicates that the inefficiencies generated by the uniform salary restriction are firm size distortions, rather than amenity mismatches.

3.1 Replacement chains

In this subsection, we develop a piece of mathematical machinery, which decomposes the change from one matching to another. A **replacement chain** comprises a sequence of workers $(w_k)_{k=0}^{N-1} \subseteq$

$\mathbf{W}$ and a sequence of firms $(F_k)_{k=0}^{N} \subseteq \mathbf{F} \cup \{\emptyset\}$ such that no worker is repeated:

$$\forall k \neq j : w_k \neq w_j,$$

and no adjacent firms are the same:

$$\forall k \leq N-1 : F_k \neq F_{k+1}.$$

A replacement chains alters a matching by moving each worker to the subsequent firm. Since workers are interchangeable in production, replacement chains allow us to consider large changes to matchings in which at most only the production of the first and last firm is affected.

Consider a matching μ and replacement chain $\chi = \left((w_k)_{k=0}^{N-1}, (F_k)_{k=0}^{N}\right)$ such that each $w_k \in \mu(F_k)$. We define $\mu + \chi$ as the matching constructed by moving each worker w_k from F_k to F_{k+1}:

$$\left(\mu + \chi\right)(w) = \begin{cases} F_{k+1} & \text{if } w = w_k; \\ \mu(w) & \text{if } w \notin (w_k)_{k=0}^{N-1}, \end{cases}$$

and we say that χ is a replacement chain from μ to another matching μ' if

$$\forall k \in 0, \ldots, N-1 : w_k \in \mu(F_k) \cap \mu'(F_{k+1}).$$

If $\mu \neq \mu'$, there exists some replacement chain from μ to μ'.[8] Note that if χ is a replacement chain from μ to μ', it need not be the case that $\mu + \chi = \mu'$. Rather, it is necessarily the case that there exists a sequence of replacement chains $\chi_1, \chi_2, \ldots, \chi_k$, all from μ to μ', such that $\mu + \chi_1 + \chi_2 + \ldots + \chi_k = \mu'$.

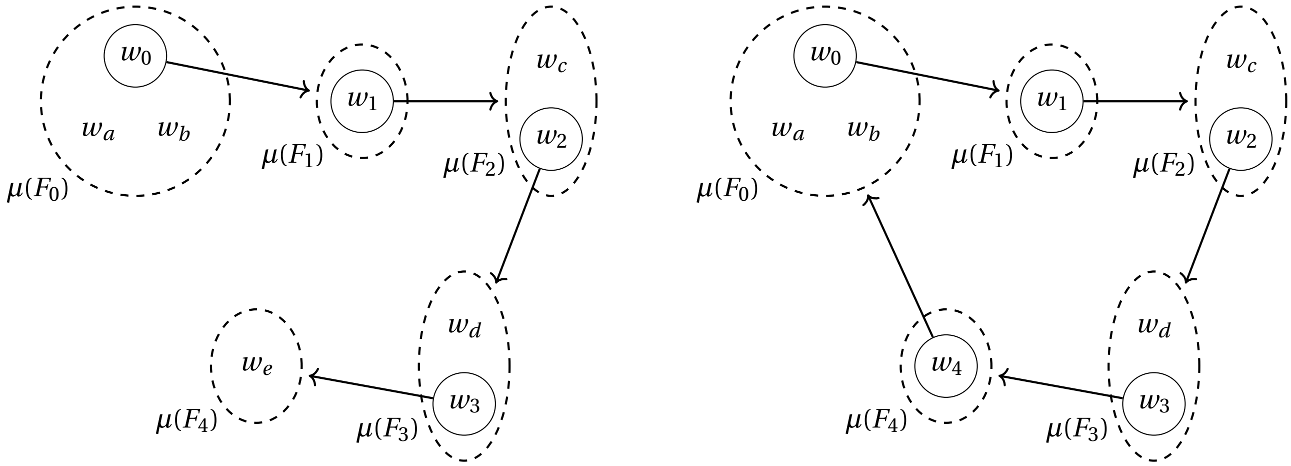
Consider some replacement chain $\chi = \left((w_k)_{k=0}^{N-1}, (F_k)_{k=0}^{N}\right)$. χ is **cyclic** if $F_0 = F_N$. χ is **acyclic** if $F_0 \neq F_N$.

Two replacement chains are depicted in Figure 3. The replacement chain in Panel (a) moves worker w_0 from firm F_0 to firm F_1, moves w_1 from F_1 to F_2, moves w_2 from F_2 to F_3, and moves w_3 from F_3 to F_4. It is acyclic because it starts and ends at different firms. The replacement chain in Panel (b) is identical, except that it additionally moves w_4 from F_4 to F_0, and is thus cyclic.

Replacement chains are useful because, from any inefficient matching, there exists a replacement chain that increases value:

Lemma 2. *Let μ and μ^* be matchings such that $value\left(\mu^*\right) > value\left(\mu\right)$. There exists a replacement chain χ from μ to μ^* such that $value\left(\mu + \chi\right) > value\left(\mu\right)$. Moreover, for each firm F: $\left|\left(\mu + \chi\right)(F)\right| \leq \max\left\{\left|\mu(F)\right|, \left|\mu^*(F)\right|\right\}$.*

[8] For example, taking any worker w such that $\mu(w) \neq \mu'(w)$, then the trivial replacement chain $\left((w), \left(\mu(w), \mu'(w)\right)\right)$ is a replacement chain from μ to μ'.



(a) An acyclic replacement chain

(b) A cyclic replacement chain

Figure 3: Two replacement chains

Lemma 2 follows from our assumption that firms treat workers as gross substitutes. For example, consider two matchings μ, μ^* such that value$\left(\mu^*\right) >$ value$\left(\mu\right)$, and μ and μ^* only differ for two workers w_1, w_2, who are both matched to firm F_1 in μ and F_2 in μ^*:

$$\{w_1, w_2\} = \mu(F_1) \cap \mu^*(F_2).$$

There is no replacement chain χ such that $\mu^* = \mu + \chi$. However, given gross substitutes, if moving *both* w_1 and w_2 from F_1 to F_2 increases value, then either moving w_1 from F_1 to F_2 increases value, or moving w_2 from F_1 to F_2 increases value (or both).

The proof of Lemma 2 generalizes the above intuition. We present an algorithm that necessarily finds the required replacement chain. This replacement chain does not grow any firm to be bigger than its size in μ^*, and so for each firm F, $\left|\left(\mu + \chi\right)(F)\right| \leq \max\left\{\left|\mu(F)\right|, \left|\mu^*(F)\right|\right\}$.

We next show that only acyclic replacement chains can improve a stable outcome:

Lemma 3. *Let $\left(\mu, s\right)$ be a stable outcome. There exists no cyclic replacement chain χ such that value$\left(\mu + \chi\right) >$ value$\left(\mu\right)$.*

Given that firm production depends only on the number of workers each firm employs, cyclic replacement chains do not change production. Thus, if a cyclic replacement chain increases an outcome's value, it must do so by rearranging workers to increase their total amenities. We show that some worker must have higher utility after the rearrangement, violating No Envy and so (by Proposition 1) violating the assumed stability of the outcome.

3.2 Relationships between stability and efficiency

In this subsection we sketch the proof of Proposition 2. Proposition 1 told us that an outcome with No Envy, No Firing, and No Poaching is stable. As such, to prove that outcomes with both No Envy and Marginal Product Salaries are stable we can prove this lemma:.

Lemma 4. *An outcome with Marginal Product Salaries has both No Firing and No Poaching.*

The No Firing condition is equivalent to the requirement that each firm F's salary be no greater than $\Delta^-_\mu(F)$. If firm F's salary is at least $\Delta^+_\mu(F)$, then any higher salary would be strictly greater than $\Delta^+_\mu(F)$, and thus at such a salary the firm would be worse off hiring an additional worker. Thus, $s(F) \geq \Delta^+_\mu(F)$ implies that (μ, s) has No Poaching.

To complete the proof of the first claim of Proposition 2, we use replacement chains to show that *any* stable outcome with Marginal Product Salaries is efficient. Consider first an outcome (μ, s) such that the value of μ can be increased by moving one worker w from F to F':

$$\Delta^+_\mu(F') - \Delta^-_\mu(F) + \alpha_w(F') - \alpha_w(F) > 0.$$

If (μ, s) has Marginal Product Salaries, then $s(F') \geq \Delta^+_\mu(F')$ and $s(F) \leq \Delta^-_\mu(F)$, and thus:

$$s(F') - s(F) + \alpha_w(F') - \alpha_w(F) > 0,$$

which shows that (μ, s) does not have No Envy, and so by Proposition 1 is not a stable outcome.

An inefficient outcome cannot necessarily be improved by moving only one worker. However, Lemmas 2 and 3 guarantee the existence of an acyclic replacement chain with which the above argument can be extended to any inefficient outcome. Formalizing this logic is how we complete the proof that *any* stable outcome with Marginal Product Salaries is efficient.

To prove the second claim of Proposition 2, we must show that one cannot improve the value of a stable outcome by reallocating workers (keeping firm sizes unchanged). The matching in a stable outcome can be transformed into a matching with hedonic efficiency using a sequence of cyclic replacement chains. Lemma 3 tells us that stable outcomes never have a value-improving cyclic replacement chain. Thus the the stable outcome must already be hedonic efficient.

3.3 An efficient stable outcome exists

We now sketch our proof of Theorem 1. Given an efficient matching, we construct a salary schedule that has both No Envy and Marginal Product Salaries, and so by Proposition 2 is stable.

Consider any matching μ. For any two firms $F, G \in \mathbf{F} \cup \{\emptyset\}$ such that $\mu(G) \neq \emptyset$, let $d^1_\mu(F, G)$ be the gap between their salaries that ensures that none of G's workers strictly prefer working at F:

$$d^1_\mu(F, G) \equiv \max_{w \in \mu(G)} \{\alpha_w(F) - \alpha_w(G)\}.$$

For every firm F define $d^1_\mu(F,F) = 0$ and for unmatched firms $G \neq F$ with $\mu(G) = \emptyset$, define $d^1_\mu(F,G) = -\infty$. For an outcome (μ, s) to have No Envy, it is necessary that no worker at firm G prefers working at firm F: *i.e.*, $s(F) \leq s(G) - d^1_\mu(F,G)$.

Let us now consider the maximal level of firm F's salary consistent with the outcome (μ, s) having No Envy, treating firm G's salary $s(G)$ as fixed. We require $s(F) \leq s(G) - d^1_\mu(F,G)$. However, if there is a third firm H, we also require that $s(F) \leq s(H) - d^1_\mu(F,H)$, and that $s(H) \leq s(G) - d^1_\mu(H,G)$. Combining inequalities, this requires that $s(F) \leq s(G) - \left(d^1_\mu(F,H) + d^1_\mu(H,G)\right)$. To capture the idea that $s(F)$ must simultaneously satisfy both the inequalities above, we define the object d^∞_μ:

$$d^\infty_\mu(F,G) \equiv \max_{N,\ (F_k)_{k=1}^N} \left\{ \sum_{k=1}^{N-1} d^1_\mu(F_k, F_{k+1}) \right\} \text{ such that } F_1 = F \text{ and } F_N = G.$$

The function d^∞_μ optimizes over all lists of firms $(F_k)_{k=1}^N$; these lists can include a firm more than once and thus are arbitrarily long. Nonetheless, Lemma 5 will ensure that d^∞_μ is well-defined, provided that μ is hedonic efficient. Lemma 5 tells us that, if a candidate list contains duplicates, there must be a list without duplicates for which the sum $\sum_{k=1}^{N-1} d^1_\mu(F_k, F_{k+1})$ is at least as large. Thus, when constructing d^∞_μ, we can ignore any list with duplicates.

Lemma 5. *Let μ be hedonic efficient. For any firms $(F_k)_{k=1}^N$ with $F_1 = F_N$: $\sum_{k=1}^{N-1} d^1_\mu(F_k, F_{k+1}) \leq 0$.*

Let us now consider an efficient matching μ^*. Our candidate salary schedule s^* is:

$$s^*(F) \equiv \min_{G \in \mathbf{F} \cup \{\emptyset\}} \left\{ \Delta^-_{\mu^*}(G) - d^\infty_{\mu^*}(F,G) \right\}. \tag{1}$$

Given that μ^* is efficient, it must be hedonic efficient, and so Lemma 5 ensures that $d^\infty_{\mu^*}$ is well-defined.[9] The salary schedule s^* sets salaries as high as is possible while ensuring that the outcome (μ^*, s^*) has both No Envy and No Firing.

By Proposition 2, the following lemma implies Theorem 1.

Lemma 6. *The outcome (μ^*, s^*) has No Envy and Marginal Product Salaries.*

That (μ^*, s^*) has No Envy follows from the definition of $d^\infty_{\mu^*}$. Marginal Product Salaries requires that, for each firm F, $s^*(F) \in \left[\Delta^+_{\mu^*}(F), \Delta^-_{\mu^*}(F)\right]$. The upper bound $s^*(F) \leq \Delta^-_{\mu^*}(F)$ follows immediately from the definition of the salary schedule. The lower bound $s^*(F) \geq \Delta^+_{\mu^*}(F)$ follows from the efficiency of μ^*: the proof of Lemma 6 shows how one could increase the value of μ^* were it the case that, for some firm F, $s^*(F) < \Delta^+_{\mu^*}(F)$.

[9] The terms $d^\infty_{\mu^*}$ can also be computed easily. They are the maximal weighted paths in the directed graph with nodes $\mathbf{F} \cup \{\emptyset\}$ and weights $d^1_\mu(F,G)$. They can be thus calculated in polynomial time using the Floyd–Warshall algorithm (Cormen, Leiserson, Rivest, & Stein, 2022).

4 Necessary Conditions for Labor Market Inefficiencies

We have shown that the uniform salary restriction *can* permit inefficient stable outcomes. In this section, we explore more directly *why* these inefficiencies arise. We first demonstrate exactly how the uniform salary restriction can stabilize inefficient outcomes:

Proposition 3. *Consider an inefficient stable outcome* (μ, s). *There exists a salary* s', *a firm* F, *and a worker* w *such that* $s' < \Delta_{\mu}^{+}(F)$, *and* w *strictly prefers to work for firm* F *at salary* s' *than for firm* $\mu(w)$ *at salary* $s\big(\mu(w)\big)$.

To prove Proposition 3, we again invoke Lemmas 2 and 3. These results tells us that, for every inefficient matching, there must be an acyclic, value-increasing replacement chain from that matching to the efficient matching. Because the replacement chain is acyclic, it increases the size of the last firm. Because the replacement chain is value-increasing, the marginal product of moving the last worker to the last firm must be greater than the salary needed to induce the worker to move.

Proposition 3 tells us that, in every inefficient stable outcome, some worker is willing to move to a new firm for a salary less than what that firm would gain from hiring her. The firm refuses to hire her, however, because doing so would require increasing the pay of its existing workers.

In the remainder of this section, we consider two conditions that ensure no such worker can exist. We say that firm F has **common value amenities** if every worker receives the same amenity from working at F:

$$\forall\, w, w' \in \mathbf{W}:\ \alpha_w(F) = \alpha_{w'}(F).$$

Proposition 4. *If every firm has common value amenities, then every stable outcome is efficient.*

By Proposition 1, stable outcomes have No Envy. When all firms have common value amenities, the No Envy condition equalizes workers' utilities. To prove Proposition 4, we use this structure within a replacement chain to construct a blocking coalition for any inefficient stable outcome.

Proposition 4 exposes one source of inefficiency: Firms want to lower their salaries to price out 'expensive' workers, even when employing those workers would be efficient. This incentive does not arise when firms have common value amenities since no worker is relatively more expensive than any other.

We say that two firms $F' \neq F$ are **duplicates** if

$$\forall\, N \in \mathbb{N}: y_F(N) = y_{F'}(N),$$
$$\text{and } \forall\, w \in \mathbf{W}: \alpha_w(F) = \alpha_w(F').$$

Proposition 5. *If every firm has a duplicate, then every stable outcome is efficient.*

To prove Proposition 5, we first note that, by No Envy, if two firms are duplicates, then they must pay the same salary. Thus, any firm could poach any worker from its duplicate by paying an infinitesimally higher salary. No Poaching then implies that each firm F's salary be at least equal to their increased output from hiring another worker (i.e., $\Delta^+_\mu(F)$). In addition, No Firing requires that each firm F's salary be no greater than the decrease in output from firing a marginal worker (i.e., $\Delta^-_\mu(F)$). The outcome thus has Marginal Product Salaries and so by Proposition 2 is efficient.

Proposition 5 exposes another source of inefficiency: Firms can pay salaries beneath their marginal products only if they are differentiated.[10] Recall that inefficiencies only arise because of the gap between Marginal Product Salaries and the conjunction of No Firing and No Poaching. The duplicate firm setting allows firms to profitably poach workers unless they pay their marginal product, eliminating this gap.

This section has shown how labor market inefficiencies require both the uniform salary restriction and two-sided heterogeneity — workers having heterogeneous preferences over heterogeneous firms.

5 Worker and Firm Welfare across Stable Outcomes

Let the partial order $\succeq_{\mathbf{W}}$ represent workers' unanimous preferences across outcomes:

$$\left(\mu, s\right) \succeq_{\mathbf{W}} \left(\mu', s'\right) \iff \forall\, w : u_w\left(\mu(w), s(\mu(w))\right) \geq u_w\left(\mu'(w), s'(\mu'(w))\right),$$

and let the partial order $\succeq_{\mathbf{F}}$ represent firms' preferences:

$$\left(\mu, s\right) \succeq_{\mathbf{F}} \left(\mu', s'\right) \iff \forall F : \pi_F\left(\left|\mu(F)\right|, s(F)\right) \geq \pi_F\left(\left|\mu'(F)\right|, s'(F)\right).$$

In this section, we show that there exists one efficient stable that pays the highest salaries, is globally optimal for workers and globally pessimal for firms:

Theorem 2. *There exists a stable outcome $\left(\mu^*, s^*\right)$ such that, in comparison to any other stable outcome $\left(\mu, s\right)$:*

1. *it is more efficient: value$\left(\mu^*\right) \geq$ value$\left(\mu\right)$,*
2. *it has greater salaries: $s^* \geqq s$,*

[10] In Appendix E, we explore when each *worker* has a duplicate. In contrast to the results above, duplicating workers has no effect on the set of stable outcomes, provided that production functions are 'stretched' appropriately.

3. *it is preferred by workers:* $(\mu^*, s^*) \succeq_{\mathbf{W}} (\mu, s)$*, and*
4. *it is less preferred by firms:* $(\mu, s) \succeq_{\mathbf{F}} (\mu^*, s^*)$.

Let (μ^*, s^*) be the efficient outcome constructed in Section 3. We prove Theorem 2 by showing that (μ^*, s^*) satisfies the theorem's four claims relative to every other stable outcome. Thus, this theorem tells us that (μ^*, s^*) not only exists, but has unique and important properties.

We also characterize worker and firm welfare across stable outcomes more generally:

Proposition 6. *For any stable outcomes* (μ, s) *and* (μ', s')*:*

1. $(\mu, s) \succeq_{\mathbf{W}} (\mu', s')$ *if and only if* $s \geqq s'$*;*
2. *if* $(\mu, s) \succeq_{\mathbf{W}} (\mu', s')$*, then* $(\mu', s') \succeq_{\mathbf{F}} (\mu, s)$.

The first claim of Proposition 6 tells us that salaries are globally aligned with worker welfare. The second claim of Proposition 6 tells us that a stable outcome which is better for all workers will be worse for all firms. Surprisingly, the converse of the second claim does not hold, violating the dual lattice structure typical of two-sided matching models.

5.1 Worker welfare, firm welfare, and salaries

In this subsection, we sketch the proof of Proposition 6.

If one outcome has greater salaries than another, a worker matched to the same firm in both outcomes necessarily prefers the former outcome over the latter. Of course, workers who switch firms between the two outcomes could prefer either outcome. However, if both outcomes are stable, No Envy guarantees that each worker prefers the firm she is matched to over any other firm, given that firm's salaries. By combining inequalities, we prove the first claim of Proposition 6: if one *stable* outcome has greater salaries than another, then all workers must prefer the former outcome over the latter.

If one stable outcome is preferred by all workers over another, then no firm could strictly prefer the former outcome to the latter: if they did, they could block the latter outcome in a coalition with the workers to which they are matched in the former. This proves the second claim of Proposition 6.

The converse of the second claim of Proposition 6 does not hold: Given two stable outcomes, one can be better for all firms and some of the workers. In many-to-one matching, a worker may be unable to form a blocking coalition with only a firm: she might need the support of her fellow workers as well. We require that each firm pays all its workers the same salary, and thus the salary sufficiently generous to earn the support of her fellow workers may cost her the support of the firm. Example C.1, in Appendix C, illustrates this phenomenon.

5.2 A worker-optimal stable outcome

In this subsection, we sketch the proof of Theorem 2. We do so by showing that the outcome $\left(\mu^*, s^*\right)$ constructed in Section 3 satisfies all four claims of Theorem 2 relative to any other stable outcome; the first claim follows immediately from its construction.

We next show that the outcome $\left(\mu^*, s^*\right)$ sets salaries maximally: for any firm F_1, there is some reason why that $s^*(F_1)$ cannot be increased. One situation is that $s^*(F_1) = \Delta^-_{\mu^*}(F_1)$ — *i.e.*, the firm cannot increase its salary without breaching the No Firing constraint. The other situation is that F_1 is the first in a sequence of firms with binding No Envy constraints; these constraints prevent each firm in the sequence from increasing its salary without also increasing the salary of the next. The sequence concludes at a firm F_N which has salary $s^*(F_N) = \Delta^-_{\mu^*}(F_N)$ and which thus cannot have a higher salary without breaching the No Firing constraint. In summary:

Lemma 7. *Let $\left(\mu^*, s^*\right)$ be the outcome constructed in Section 3. For each firm F_1, either $s^*(F_1) = \Delta^-_{\mu^*}(F_1)$, or there exists a list of firms $(F_k)_{k=2}^N$ each containing some worker $w_k \in \mu^*(F_k)$, such that each worker w_k is indifferent between firm F_k and firm F_{k-1}:*

$$\alpha_{w_k}(F_{k-1}) + s^*(F_{k-1}) = \alpha_{w_k}(F_k) + s^*(F_k),$$

and $s^*(F_N) = \Delta^-_{\mu^*}(F_N)$.

Lemma 7 suggests that $\left(\mu^*, s^*\right)$ has the highest salaries possible for an efficient stable outcome. However, it is not obvious whether other, inefficient stable outcomes could have higher salaries for at least some workers. Given that firms' production functions have decreasing differences, one might worry that an outcome in which a firm is matched to inefficiently few workers would have higher salaries than an efficient outcome.

To prove the second claim of Theorem 2, we show such an outcome cannot be stable. For example, consider an outcome (μ, s), identical to $\left(\mu^*, s^*\right)$ except that some firm F is both matched to fewer workers $\mu(F) \subset \mu^*(F)$ and pays a higher salary $s(F) > s^*(F)$. Such a salary would make the displaced workers $\mu^*(F) \setminus \mu(F)$ envious, destabilizing the outcome. Our proof generalizes this intuition.

Combining the second claim of Theorem 2 with Proposition 6, we can conclude that $\left(\mu^*, s^*\right)$ is simultaneously the worker-optimal and firm-pessimal stable outcome, completing the proof of Theorem 2.

6 Designing a Centralized Labor Market

This paper has shown that uniform salaries can distort labor market efficiency. In this section, we ask whether a centralized mechanism can eliminate this distortion. We will show that such a mechanism can elicit workers' preferences, but not firms' production functions:

Theorem 3.

1. *When firms' production functions are public information, a strategyproof mechanism can elicit workers' preferences to implement an efficient stable outcome.*
2. *When firms' production functions are private information, no strategyproof mechanism can generally implement an efficient stable outcome.*

The second claim of Theorem 3 can be proved with the following example.

Example 2. $\mathbf{F} = \{F\}$. $y_F(N) = \beta N$; $\beta \in \{1, 6\}$. $\mathbf{W} = \{w_1, w_2\}$. $\alpha_{w_1}(F) = 0$. $\alpha_{w_2}(F) = -4$.

Example 2 generalizes Example 1. When $\beta = 6$, Example 2 is identical to Example 1, and the efficient matching $\mu^6(w_1) = \mu^6(w_2) = F$ is supported by a salary $s^6(F) \in [4, 6]$. When $\beta = 1$, the efficient matching is $\mu^1(w_1) = F; \mu^1(w_2) = \emptyset$. This matching is supported by a salary $s^1(F) \in [0, 1]$.

Consider the mechanism design problem of implementing $\left(\mu^6, s^6\right)$ when $\beta = 6$ and $\left(\mu^1, s^1\right)$ when $\beta = 1$, where only the firm knows β. By the revelation principle, we can consider only mechanisms in which the firm reports its type. If it reports $\beta = 1$, it will be matched to one worker, pay salary $s^1(F) \in [0, 1]$, and thus receive profit $\beta - s^1(F) \geq \beta - 1$. If it reports $\beta = 6$, it will be matched to two workers, pay salary $s^6(F) \in [4, 6]$, and thus receive profit $2\left(\beta - s^6(F)\right) \leq 2(\beta - 4)$. In particular, when the true value of the firm's productivity is $\beta = 6$, it would receive at least profit 5 from reporting $\beta = 1$, while it would receive at most profit 4 from reporting $\beta = 6$. It will thus not report truthfully. The example thus proves the second claim of Theorem 3.

We now sketch how a mechanism *can* elicit workers' preferences to implement an efficient outcome. In particular, under a mechanism that implements the worker-optimal stable outcome constructed in Section 3,it is weakly dominant for each worker to report truthfully.

Fix all workers' reports except for some worker w. Let $\left(\mu^*, s^*\right)$ be the implemented worker-optimal stable outcome (not necessarily the actual worker-optimal stable outcome) were w to report truthfully, and let $\left(\mu^\circ, s^\circ\right)$ be the implemented worker-optimal stable outcome were w to misreport and benefit, with $F^\circ = \mu^\circ(w)$.

Due to the No Envy (with respect to reported preferences) of *both* $\left(\mu^*, s^*\right)$ and $\left(\mu^\circ, s^\circ\right)$, for w to benefit from misreporting, the salary of her firm under misreporting must be higher than that firm's salary under truthful reporting: $s^\circ(F^\circ) > s^*(F^\circ)$. By Lemma 7, either the No Firing constraint $s^*(F^\circ) \leq \Delta^-_{\mu^*}(F^\circ)$ is binding, or firm F°'s No Envy constraint is binding for some worker $w' \notin \mu^*(F^\circ)$: $s^*(F^\circ) + \alpha_{w'}(F^\circ) = s^*\left(\mu^*(w')\right) + \alpha_{w'}\left(\mu^*(w')\right)$. Our proof shows that worker w's misreport can neither relax the No Envy constraints nor *increase* the marginal product at F°.

Theorem 3 echos earlier results in the two-sided matching literature: there is no mechanism that is strategyproof for both sides of the market; and implementing the worker-optimal stable outcome is strategyproof for workers but not firms (Roth & Sotomayor, 1990). Our context has the added twist that the worker-optimal stable outcome is efficient while other stable outcomes need not be. We thus provide an argument in favor of worker-optimal mechanisms: that they provide efficient scale.

7 Conclusion

For many firms, employing more workers would require paying higher salaries. In classic job matching models, these higher salaries need only be paid to the firm's new workers; the firm can leave its existing workers' salaries unchanged. In other words, these models assume that labor markets exhibit price discrimination. This paper has argued that, without price discrimination, labor markets can suffer from distortions. We introduced a tractable matching framework to model these distortions.

This matching framework has yielded new insights into labor market distortions. We showed that only firm sizes are distorted: conditional on firm sizes, the matching of workers to firms is efficient. We showed that these distortions are beneficial to firms and are harmful to workers. Further, we showed that these distortions stem from two-sided heterogeneity: when each firm has a duplicate, or when each firm's amenities are equally appreciated by all workers, every stable outcome is efficient.

We have used this framework to assess a potential solution to labor market distortions: a centralized matching mechanism. To be successful, such a mechanism would have to implement an outcome that is both efficient and stable. We showed that one efficient outcome is indeed always stable. To identify an efficient outcome, such a mechanism may also need to elicit the non-pecuniary amenities that workers receive from employment, or elicit the technologies with which firms produce output. We showed that a strategyproof mechanism can elicit the former but not the latter.

Ideally, a lighter-touch policy could implement an efficient outcome. For example, in a pure monopsony consisting of only one firm, a minimum salary can incentivize efficient employment. However, with multiple firms an efficient minimum salary must be firm-specific: a uniform minimum salary cannot efficiently allocate labor between a high amenity, low-productivity firm and a low-amenity, high-productivity firm.

While a market designer could impose firm-specific minimum salaries, doing so is not trivial. We show that stable outcomes with Marginal Product Salaries are efficient. However, to know what the right marginal products are, the market designer must know the efficient matching. Given that these minimum salaries require that the designer know an efficient matching, it seems simpler to just impose that matching directly.

Our baseline model requires that workers be interchangeable in production. When workers are not interchangeable in production, but salaries are still required to be uniform within a firm, a stable outcome may not exist. On the other hand, our results trivially extend to cover labor markets comprising multiple occupations provided that firms could set different salaries to different occupations, that each worker could only work in one occupation, and that firms' production was additively separable over occupations. Tractable models with heterogeneous worker productivity would be a valuable goal for future work. For example, our results may extend to the case in which workers productivities vary and firms choose a piece-rate, proportional to each worker's productivity. Similarly, it would be interesting to derive the minimal restrictions on worker heterogeneity under which a stable outcome with uniform salaries always exists.

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A Proofs of main results

This appendix contains proofs of our main results. The proofs of all our lemmas are contained in the Online Appendix. We order the following proofs so that they can be read linearly. Where possible, we preserve the order in which their statements appear in the main text.

Proposition 1. *An outcome is stable if and only if it has No Envy, No Firing, and No Poaching.*

Proof. We prove Proposition 1 in six steps.

Step 1: If an outcome is stable, then it has No Firing.

Proof of Step 1: Consider a firm F such that $\mu(F) \neq \emptyset$. If the outcome lacks No Firing, then the firm is making a loss on its marginal worker. It would be better off being matched to one worker less at the same salary:

$$\forall w \in \mu(F) : \pi_F\left(\left|\mu(F) \setminus \{w\}\right|, s(F)\right) > \pi_F\left(\left|\mu(F)\right|, s(F)\right).$$

If $\left|\mu(F)\right| = 1$, the left hand size of the above inequality is 0, and thus the candidate outcome is not individually rational for the firm. If $\left|\mu(F)\right| > 1$, then for any worker $w \in \mu(F)$: $\left(F, \mu(F) \setminus \{w\}, s(F)\right)$ blocks $\left(\mu, s\right)$ because firm F would be strictly better off in the blocking coalition and every worker in $\mu(F) \setminus \{w\}$ would be indifferent. Thus, an outcome without No Firing cannot be stable.

Step 2: If an outcome is stable, then it has No Envy.

Proof of Step 2: Assume towards a contradiction that $\left(\mu, s\right)$ is stable but does not have No Envy:

$$\exists w \in \mathbf{W}, F \in \mathbf{F} \cup \{\emptyset\} : u_w\left(\mu(w), s(\mu(w))\right) < u_w(F, s(F)).$$

If $F = \emptyset$, the right hand side of that inequality is 0, and thus $\left(\mu, s\right)$ is not individually rational for w. Thus, if $\left(\mu, s\right)$ is stable but lacks No Envy, then $F \neq \emptyset$.

Consider first the case where $\mu(F) \neq \emptyset$. Since workers are interchangeable in production, the firm would be indifferent switching out any worker in its employ, keeping the same salary. Thus, for any $w' \in \mu(F)$, $\left(F, \mu(F) \cup \{w\} \setminus \{w'\}, s(F)\right)$ blocks $\left(\mu, s\right)$.

Now consider the case where $\mu(F) = \emptyset$. For such a firm, $s(F) = y_F(1)$ by definition. Thus:

$$\alpha_w(F) + y_F(1) > s\left(\mu(w)\right) + \alpha_w\left(\mu(w)\right),$$

while

$$\pi_F\left(|\{w\}|, y_F(1)\right) = \pi_F\left(\left|\mu(F)\right|, s(F)\right) = 0.$$

Thus, $\left(F, \{w\}, y_F(1)\right)$ blocks $\left(\mu, s\right)$.

Step 3: If an outcome is stable, then it has No Poaching.

Proof of Step 3: Assume towards a contradiction that $\left(\mu, s\right)$ is stable but lacks No Poaching: there exists F, $s' > s(F)$, and $L \in \mathbb{N}$ with $\left|\mu(F)\right| < L \leq L_F\left(s', s\right)$ such that

$$\pi_F\left(L, s'\right) = y_F(L) - Ls' \geq \pi_F\left(\left|\mu(F)\right|, s(F)\right) = y_F\left(\left|\mu(F)\right|\right) - \left|\mu(F)\right| s(F). \tag{2}$$

If $\mu(F) = \emptyset$, then $s(F) = y_F(1)$. Thus, $s' > y_F(1)$, meaning firm F would make negative profit if matched to one worker. By Lemma 1, its production function has decreasing differences, and so firm F would make a negative profit when matched to any positive number of workers. This contradicts expression (2), which requires that there be a positive L such that $\pi_F\left(L, s'\right) \geq \pi_F\left(\left|\mu(F)\right|, s(F)\right) = 0$.

If $\mu(F) \neq \emptyset$, every worker $w \in \mu(F)$ is strictly better off being employed at salary s' rather than salary $s(F)$. By assumption, there is a size-L set of workers C, who weakly prefer being matched to F at salary s' over their current match. As $L > 0$ we can require that $\mu(F) \cap C \neq \emptyset$. Thus, there is a worker in C who strictly prefers being matched to F at salary s' over their current match. By assumption, firm F weakly prefers being matched to C at salary s' over being matched to $\mu(F)$ at salary $s(F)$. Thus, $\left(F, C, s'\right)$ blocks $\left(\mu, s\right)$.

Step 4: If an outcome has No Envy, then it is individually rational for workers.

Proof of Step 4: If an outcome has No Envy, then $\forall\, w$:

$$u_w\left(\mu(w), s(\mu(w))\right) \geq u_w\left(\emptyset, 0\right) = 0.$$

Step 5: If an outcome has No Firing, then it is individually rational for firms.

Proof of Step 5: Note that individual rationality for unmatched firms is trivial: if $\mu(F) = \emptyset$, then $\pi_F\left(\left|\mu(F)\right|, s(F)\right) = 0$.

An outcome having No Firing requires that for all matched firms F:

$$s(F) \leq y_F\left(\left|\mu(F)\right|\right) - y_F\left(\left|\mu(F)\right| - 1\right). \tag{3}$$

By Lemma 1, each firm's production function has decreasing differences, and so

$$\begin{aligned}\pi_F\left(\left|\mu(F)\right|, s(F)\right) &= \sum_{i=1}^{\left|\mu(F)\right|} \left[y_F\left(i\right) - y_F\left(i-1\right) - s(F)\right] \\ &\geq \sum_{i=1}^{\left|\mu(F)\right|} \left[y_F\left(\left|\mu(F)\right|\right) - y_F\left(\left|\mu(F)\right| - 1\right) - s(F)\right].\end{aligned}$$

Combined with inequality (3), this implies that $\pi_F\left(\left|\mu(F)\right|, s(F)\right) \geq 0$.

Step 6: If an outcome has No Envy, No Firing, and No Poaching, then there are no coalitions that block it.

Proof of Step 6: Assume towards a contradiction that the coalition $\left(F, C, s'\right)$ blocks $\left(\mu, s\right)$, an outcome with No Envy, No Firing, and No Poaching, where C is a nonempty subset of $\mathbf{W}$.

First consider the case where $s' < s(F)$. Given that $s' < s(F)$, any worker previously matched to F would be strictly worse off in the coalition. Thus, $C \cap \mu(F) = \emptyset$. Consider a worker $w \in C$. Since $\left(\mu, s\right)$ has No Envy,

$$\alpha_w\left(\mu(w)\right) + s\left(\mu(w)\right) \geq \alpha_w(F) + s(F) > \alpha_w(F) + s',$$

and thus the worker prefers the original outcome over being matched to F at salary s'. This contradicts the assumption that $\left(F, C, s'\right)$ blocks $\left(\mu, s\right)$.

Next consider the case where $s' > s(F)$. As the firm is no worse off,

$$\pi_F\left(|C|, s'\right) \geq \pi_F\left(\left|\mu(F)\right|, s(F)\right). \tag{4}$$

Given that $s' > s(F)$, inequality (4) implies that

$$\pi_F\left(|C|, s(F)\right) > \pi_F\left(\left|\mu(F)\right|, s(F)\right). \tag{5}$$

Let us now show that $|C| > \left|\mu(F)\right|$. That $|C| \neq \left|\mu(F)\right|$ follows directly from inequality (5). Assume towards a contradiction that $|C| < \left|\mu(F)\right|$. As such,

$$\pi_F\left(\left|\mu(F)\right|, s(F)\right) - \pi_F\left(|C|, s(F)\right) = \sum_{i=|C|+1}^{|\mu(F)|} \left[y_F(i) - y_F(i-1) - s(F)\right]. \tag{6}$$

By Lemma 1, the production function y_F has decreasing differences, and so for any $i \leq \left|\mu(F)\right|$: $y_F(i) - y_F(i-1) \geq \Delta_\mu^-(F)$. By No Firing, firm F is not losing money on its marginal worker: $s(F) \leq \Delta_\mu^-(F)$. Together with equation (6), these inequalities imply that

$$\pi_F\left(\left|\mu(F)\right|, s(F)\right) - \pi_F\left(|C|, s(F)\right) \geq 0,$$

which contradicts inequality (5). Thus $|C| > |\mu(F)|$.

As all workers in C are no worse off in the coalition $\left(F, C, s'\right)$, it must be the case that $|C| \leq L_F\left(s', s\right)$.

Inequality (4), the fact that $|C| > |\mu(F)|$, and the fact that $|C| \leq L_F\left(s', s\right)$ together contradict the assumption that $\left(\mu, s\right)$ has No Poaching.

Finally consider the case where $s' = s(F)$. Since $\left(\mu, s\right)$ has No Envy, $\forall w \in C$:

$$\alpha_w\left(\mu(w)\right) + s\left(\mu(w)\right) \geq \alpha_w(F) + s(F) = \alpha_w(F) + s',$$

and thus workers in C are at best indifferent between the original outcome and being matched to F at salary s'. Thus, firm F must be strictly better off in the coalition. Thus, the firm would also be strictly better off being matched to C at salary slightly higher than s'. Thus, the $s' = s(F)$ case reduces to the $s' > s(F)$ considered above. □

Proposition 2.

1. *If an outcome has No Envy and Marginal Product Salaries, then it is stable and efficient.*
2. *If an outcome is stable, then it is hedonic efficient.*

Proof. We prove each claim of the proposition separately — each in two steps.

Step 1.1: If an outcome has No Envy and Marginal Product Salaries, then it is stable.

Proof of Step 1.1: Let (μ, s) be an outcome with No Envy and Marginal Product Salaries. By Lemma 4, (μ, s) has No Firing and No Poaching, and thus, by Proposition 1, it is stable.

Step 1.2: If an outcome has No Envy and Marginal Product Salaries, then it is efficient.

Proof of Step 1.2: Let (μ, s) be an outcome with *No Envy* and *Marginal Product Salaries.* By Step 1.1, (μ, s) is stable.

By Lemma 2, if μ is not efficient, there exists a replacement chain $\chi = \left((w_k)_{k=0}^{N-1}, (F_k)_{k=0}^{N}\right)$ such that value$(\mu + \chi) >$ value(μ). By Lemma 3 and the fact that (μ, s) is stable, χ is acyclic. It follows that

$$\text{value}\left(\mu+\chi\right) - \text{value}\left(\mu\right) = \Delta_\mu^+(F_N) - \Delta_\mu^-(F_0) + \sum_{k=0}^{N-1}\left[\alpha_{w_k}(F_{k+1}) - \alpha_{w_k}(F_k)\right]. \tag{7}$$

By Proposition 1, (μ, s) has No Envy. As such, for each $k = 0, \ldots, N-1$: $\alpha_{w_k}(F_{k+1}) - \alpha_{w_k}(F_k) \leq s(F_k) - s(F_{k+1})$. Thus

$$\sum_{k=0}^{N-1}\left[\alpha_{w_k}(F_{k+1}) - \alpha_{w_k}(F_k)\right] \leq \sum_{k=0}^{N-1}\left[s(F_k) - s(F_{k+1})\right] = s(F_0) - s(F_N). \tag{8}$$

By Marginal Product Salaries: $s(F_N) \geq \Delta_\mu^+(F_N)$ and $s(F_0) \leq \Delta_\mu^-(F_0)$. Combining these inequalities with equation (7) and inequality (8) yields

$$\text{value}\left(\mu+\chi\right) - \text{value}\left(\mu\right) \leq 0.$$

This contradicts our earlier claim that value$(\mu+\chi) >$ value(μ).

Two replacement chains $\chi_A = \left((w_k)_{k=0}^{N_A-1}, (F_k)_{k=0}^{N_A}\right)$ and $\chi_B = \left((w_l)_{l=0}^{N_B-1}, (F_l)_{l=0}^{N_B}\right)$ are **worker-disjoint** if they share no worker: $\{w_k\}_{k=0}^{N_A-1} \cap \{w_l\}_{l=0}^{N_B-1} = \emptyset$.

Step 2.1: If χ_A and χ_B are worker-disjoint cyclic replacement chains from μ to μ', then

$$\text{value}\left(\mu+\chi_A+\chi_B\right) = \text{value}\left(\mu+\chi_A\right) + \text{value}\left(\mu+\chi_B\right) - \text{value}\left(\mu\right).$$

Proof of Step 2.1: Note that $(\mu+\chi_A+\chi_B)$ and $(\mu+\chi_B+\chi_A)$ are both well-defined since each replacement chain moves a worker from μ to μ' not yet moved by the other chain.

Let $\chi_A = \left((w_k)_{k=0}^{N_A-1}, (F_k)_{k=0}^{N_A}\right)$ and let $\chi_B = \left((w_l)_{l=0}^{N_B-1}, (F_l))_{l=0}^{N_B}\right)$. Since χ_A and χ_B are both cyclic replacement chains from μ to μ':

$$\text{value}\left(\mu+\chi_A\right) - \text{value}\left(\mu\right) = \sum_{w\in(w_k)_{k=0}^{N_A-1}}\left[\alpha_w\left(\mu'(w)\right) - \alpha_w\left(\mu(w)\right)\right];$$

$$\text{value}\left(\mu+\chi_B\right) - \text{value}\left(\mu\right) = \sum_{w\in(w_l)_{l=0}^{N_B-1}}\left[\alpha_w\left(\mu'(w)\right) - \alpha_w\left(\mu(w)\right)\right];$$

$$\text{value}\left(\mu+\chi_A+\chi_B\right) - \text{value}\left(\mu\right) = \sum_{w\in(w_k)_{k=0}^{N_A-1}\cup(w_l)_{l=0}^{N_B-1}}\left[\alpha_w\left(\mu'(w)\right) - \alpha_w\left(\mu(w)\right)\right].$$

Since χ_A and χ_B are worker-disjoint, $(w_k)_{k=0}^{N_A-1} \cap (w_l)_{l=0}^{N_B-1} = \emptyset$, which implies that:

$$\sum_{w \in (w_k)_{k=0}^{N_A-1} \cup (w_l)_{l=0}^{N_B-1}} \left[\alpha_w\left(\mu'(w)\right) - \alpha_w\left(\mu(w)\right)\right]$$
$$= \sum_{w \in (w_k)_{k=0}^{N_A-1}} \left[\alpha_w\left(\mu'(w)\right) - \alpha_w\left(\mu(w)\right)\right] + \sum_{w \in (w_l)_{l=0}^{N_B-1}} \left[\alpha_w\left(\mu'(w)\right) - \alpha_w\left(\mu(w)\right)\right],$$

Combining the above expressions implies that

$$\text{value}\left(\mu + \chi_A + \chi_B\right) - \text{value}\left(\mu\right) = \text{value}\left(\mu + \chi_A\right) + \text{value}\left(\mu + \chi_B\right) - 2 \cdot \text{value}\left(\mu\right).$$

Thus, $\text{value}\left(\mu + \chi_A + \chi_B\right) = \text{value}\left(\mu + \chi_A\right) + \text{value}\left(\mu + \chi_B\right) - \text{value}\left(\mu\right)$.

Step 2.2: If an outcome is stable, then it is hedonic efficient.

Proof of Step 2.2: Assume towards a contradiction that $\left(\mu^\circ, s^\circ\right)$ is a stable outcome and μ° is not hedonic efficient. Let

$$\mu^* \in \underset{\mu \text{ s.t.} \forall F: |\mu(F)| = |\mu^\circ(F)|}{\arg\max} \left\{\sum_{w \in \mathbf{W}} \alpha_w\left(\mu(w)\right)\right\}$$

be a matching with the same firm sizes as μ°, but which is hedonic efficient.

Select an arbitrary worker w_0 for whom $\mu^\circ(w_0) \neq \mu^*(w_0)$; let $F_0 = \mu^\circ(w_0)$ and let $F_1 = \mu^*(w_0)$. For every firm F: $\left|\mu^\circ(F)\right| = \left|\mu^*(F)\right|$. Thus, there must be some worker $w_1 \in \mu^\circ(F_1)$ such that $\mu^\circ(w_1) \neq \mu^*(w_1)$. We can iteratively continue to identify new worker-firm pairs w_j, F_j such that $w_j \in \mu^\circ(F_j) \cap \mu^*(F_{j+1})$. Because the number of firms is, finite we must eventually find a firm F_N such that $F_N = F_i$ with $i < N$. We have constructed the cyclic replacement chain $\chi_1 = \left(\left(w_j\right)_{j=i}^{N-1}, \left(F_j\right)_{j=i}^{N}\right)$. Now repeat the above process to find a sequence of cyclic worker-disjoint replacement chains $\{\chi_m\}_{m=1}^{M}$ from μ° to μ^* such that $\left(\mu^\circ + \chi_1 + \chi_2 + \ldots + \chi_M\right) = \mu^*$. Thus

$$\text{value}\left(\mu^*\right) = \text{value}\left(\mu^\circ + \chi_1 + \chi_2 + \ldots + \chi_M\right). \tag{9}$$

Iterating Step 2.1 implies that

$$\text{value}\left(\mu^\circ + \chi_1 + \chi_2 + \ldots + \chi_M\right) = \text{value}\left(\mu^\circ + \chi_1\right) + \text{value}\left(\mu^\circ + \chi_2\right) + \ldots + \text{value}\left(\mu^\circ + \chi_M\right) - (M-1) \cdot \text{value}\left(\mu^\circ\right). \tag{10}$$

Each χ_m is cyclic and thus, by Lemma 3 , for all $m \in \{1, \ldots, M\}$: $\text{value}\left(\mu^\circ + \chi_m\right) \leq \text{value}\left(\mu^\circ\right)$. With equations (9) and (10), this implies that

$$\text{value}\left(\mu^*\right) \leq M \cdot \text{value}\left(\mu^\circ\right) - (M-1) \cdot \text{value}\left(\mu^\circ\right) = \text{value}\left(\mu^\circ\right).$$

This contradicts the assumption that $\text{value}\left(\mu^\circ\right) < \text{value}\left(\mu^*\right)$. □

Proposition 4. *If every firm has common value amenities, then every stable outcome is efficient.*

Proof. Let every firm F have common value amenity $\alpha(F)$:

$$\forall w \in \mathbf{W}: \alpha_w(F) = \alpha(F),$$

and assume towards a contradiction that there exists a stable outcome (μ, s), where μ is inefficient.

First, note that it follows from the stability of (μ, s) and Proposition 1 that (μ, s) has No Envy. In turn, No Envy implies that:

$$\forall F, F' \in \mathbf{F} \text{ such that } \mu(F) \neq \emptyset: \alpha(F) + s(F) \geq \alpha(F') + s(F'). \tag{11}$$

By Lemma 2, there exists a replacement chain $\chi = \left((w_k)_{k=0}^{N-1}, (F_k)_{k=0}^{N}\right)$ from μ to an efficient matching such that value$\left(\mu + \chi\right) >$ value$\left(\mu\right)$. By Lemma 3, χ is acyclic, and thus:

$$\begin{aligned} \text{value}\left(\mu+\chi\right) - \text{value}\left(\mu\right) &= \Delta_\mu^+(F_N) - \Delta_\mu^-(F_0) + \sum_{k=0}^{N-1} \left[\alpha(F_{k+1}) - \alpha(F_k)\right] \\ &= \Delta_\mu^+(F_N) - \Delta_\mu^-(F_0) + \alpha(F_N) - \alpha(F_0). \end{aligned}$$

Given that value$\left(\mu + \chi\right) >$ value$\left(\mu\right)$, this implies that

$$\alpha(F_0) - \alpha(F_N) < \Delta_\mu^+(F_N) - \Delta_\mu^-(F_0). \tag{12}$$

By the definition of a replacement chain, $\mu(F_0) \neq \emptyset$. Thus, by expression (11), $s(F_N) - s(F_0) \leq \alpha(F_0) - \alpha(F_N)$. With inequality (12), this implies that

$$s(F_N) - s(F_0) < \Delta_\mu^+(F_N) - \Delta_\mu^-(F_0). \tag{13}$$

Since (μ, s) is stable, Proposition 1 tells us that it has No Firing. As such, $s(F_0) \leq \Delta_\mu^-(F_0)$. With inequality (13) this implies that $s(F_N) < \Delta_\mu^+(F_N)$. Thus, firm F_N would strictly benefit from hiring an additional worker at its current salary.

If $\mu(F_N) \neq \emptyset$, it follows from (11) that

$$\forall F \in \mathbf{F}: \alpha(F_N) + s(F_N) \geq \alpha(F) + s(F).$$

By the definition of an acyclic replacement chain, $w_0 \notin \mu(F_N)$. Thus, worker w_0 would be willing to work at firm F_N at salary $s(F_N)$. Thus, the coalition $\left(F_N, \mu(F_N) \cup \{w_0\}, s(F_N)\right)$ would block (μ, s), contradicting (μ, s) being stable. Thus, it must be the case that $\mu(F_N) = \emptyset$.

However, if $\mu(F_N) = \emptyset$, then F_N must be making zero profit. It could thus offer to employ worker w_0 at salary $\Delta_\mu^+(F_N)$ and still make zero profit. By inequality (12), $\alpha(F_0) + \Delta_\mu^-(F_0) < \alpha(F_N) + \Delta_\mu^+(F_N)$. Recall that $s(F_0) \leq \Delta_\mu^-(F_0)$. Thus, worker w_0 would be strictly better off. The coalition $\left(F_N, \{w_0\}, \Delta_\mu^+(F_N)\right)$ blocks (μ, s), contradicting (μ, s) being stable.

Given that both $\mu(F_N) \neq \emptyset$ and $\mu(F_N) = \emptyset$ yield contradictions, there can be no stable outcome (μ, s), where μ is inefficient.

$\square$

Proposition 5. *If every firm has a duplicate, then every stable outcome is efficient.*

Proof. Let (μ, s) be a stable outcome. Let F' be the duplicate of F. That (μ, s) has No Envy implies

$$\forall w \in \mu(F) : \alpha_w(F) + s(F) \geq \alpha_w(F') + s(F') = \alpha_w(F) + s(F'),$$

where the equality follows from the assumption that F, F' are duplicates. Thus: $\mu(F) \neq \emptyset \implies s(F) \geq s(F')$.

If $\mu(F) = \emptyset$ then by construction $s(F) = y_F(1)$. As F, F' are duplicates: $y_F(1) = y_{F'}(1)$. By Lemma 1, $y_{F'}$ has decreasing differences, and so $y_{F'}(1) \geq \Delta^-_\mu(F')$. By Proposition 1, (μ, s) has No Firing, which requires $\Delta^-_\mu(F') \geq s(F')$. Combining these expressions we see that $\mu(F) = \emptyset$ implies $s(F) \geq s(F')$. Given the prior paragraph, this implies $s(F) \geq s(F')$ for all duplicates F, F'. Symmetrically, $s(F') \geq s(F)$. Therefore, $s(F) = s(F')$.

If $\mu(F') \neq \emptyset$ and $s(F) < \Delta^+_\mu(F)$, then F would be strictly better off being additionally matched to $w \in \mu(F')$ at salary $s(F)$, while w would be indifferent (because $s(F) = s(F')$). Thus, $(F, \mu(F) \cup \{w\}, s(F))$ would block (μ, s), contradicting the assumption that (μ, s) is a stable outcome. Thus, if $\mu(F') \neq \emptyset$, then $s(F) \geq \Delta^+_\mu(F)$. If $\mu(F') = \emptyset$, then $s(F) = s(F') = y_{F'}(1) = y_F(1) \geq \Delta^+_\mu(F)$, with the last inequality following from y_F having decreasing differences (by Lemma 1). Thus, if $\mu(F') = \emptyset$, then $s(F) \geq \Delta^+_\mu(F)$. In summary, for all F: $s(F) \geq \Delta^+_\mu(F)$.

By (μ, s) having No Firing, $s(F) \leq \Delta^-_\mu(F)$. We have shown that for any firm F with a duplicate, $s(F) \in \left[\Delta^+_\mu(F), \Delta^-_\mu(F)\right]$. When all firms have a duplicate, this implies that (μ, s) has Marginal Product Salaries. By Proposition 2, this implies that μ is efficient. □

Proposition 6. *For any stable outcomes (μ, s) and (μ', s'):*

1. $(\mu, s) \succeq_{\mathbf{W}} (\mu', s')$ *if and only if* $s \geqq s'$;
2. *if* $(\mu, s) \succeq_{\mathbf{W}} (\mu', s')$, *then* $(\mu', s') \succeq_{\mathbf{F}} (\mu, s)$.

Proof. We prove Proposition 6 in three steps.

Step 1: For any stable outcomes (μ, s) and (μ', s'): $s \geqq s' \implies (\mu, s) \succeq_{\mathbf{W}} (\mu', s')$.

Proof of Step 1: Since (μ, s) is stable, Proposition 1 tells us that it has No Envy. As such, it must be the case that for every worker w:

$$\alpha_w\left(\mu(w)\right) + s\left(\mu(w)\right) \geq \alpha_w\left(\mu'(w)\right) + s\left(\mu'(w)\right),$$

while $s \geqq s'$ implies that $\alpha_w\left(\mu'(w)\right) + s\left(\mu'(w)\right) \geq \alpha_w\left(\mu'(w)\right) + s'\left(\mu'(w)\right)$.

Step 2: For any stable outcomes (μ, s) and (μ', s'): $(\mu, s) \succeq_{\mathbf{W}} (\mu', s') \implies s \geqq s'$.

Proof of Step 2: For every worker:

$$\alpha_w\left(\mu(w)\right) + s\left(\mu(w)\right) \geq \alpha_w\left(\mu'(w)\right) + s'\left(\mu'(w)\right) \geq \alpha_w\left(\mu(w)\right) + s'\left(\mu(w)\right),$$

where the first inequality follows from $(\mu, s) \succeq_{\mathbf{W}} (\mu', s')$, while the second follows from the fact that (μ', s') is stable and so by Proposition 1 has No Envy. This implies $s(F) \geq s'(F)$ for every firm F such that $\mu(F) \neq \emptyset$.

For firms F such that $\mu(F) = \emptyset$, $s(F) = y_F(1)$ by definition. If $\mu'(F) = \emptyset$ for such firms, then by definition $s'(F) = y_F(1) = s(F)$. If $\mu'(F) \neq \emptyset$ for such firms then $\Delta^-_{\mu'}(F) \leq y_F(1)$, because by Lemma 1 y_F has decreasing differences. By Proposition 1, (μ', s') has No Firing and so $s'(F) \leq \Delta^-_{\mu'}(F)$. Combining these inequalities with $y_F(1) = s(F)$ implies that $s'(F) \leq s(F)$.

Step 3: For any stable outcomes (μ, s) and (μ', s'): $(\mu, s) \succeq_{\mathbf{W}} (\mu', s') \implies (\mu', s') \succeq_{\mathbf{F}} (\mu, s)$.
Proof of Step 3: Assume towards a contradiction there exists a firm F such that $\pi_F\left(\left|\mu(F)\right|, s(F)\right) > \pi_F\left(\left|\mu'(F)\right|, s'(F)\right)$. Since (μ', s') is individually rational for firms, $\pi_F\left(\left|\mu'(F)\right|, s'(F)\right) \geq 0$. Thus, $\mu(F) \neq \emptyset$. By assumption, all workers in $\mu(F)$ weakly prefer (μ, s) to (μ', s'). Thus, $(F, \mu(F), s(F))$ blocks (μ', s'), contradicting the assumption that (μ', s') is a stable outcome.

□

Theorem 2. *There exists a stable outcome (μ^*, s^*) such that, in comparison to any other stable outcome (μ, s):*

1. *it is more efficient: value$(\mu^*) \geq$ value(μ),*
2. *it has greater salaries: $s^* \geqq s$,*
3. *it is preferred by workers: $(\mu^*, s^*) \succeq_{\mathbf{W}} (\mu, s)$, and*
4. *it is less preferred by firms: $(\mu, s) \succeq_{\mathbf{F}} (\mu^*, s^*)$.*

Proof. Consider the outcome (μ^*, s^*) constructed in Section 3. We prove Theorem 2 by showing that (μ^*, s^*) satisfies the theorem's four claims relative to any other stable outcome (μ, s). Claim 1 holds by definition. Given Claim 2, claims 3 and 4 follow from Proposition 6. It thus remains to prove claim 2, which we do in seven steps.

Let $\mathscr{I} = \{F : s(F) > s^*(F)\}$.

Step 1: $\forall\, w \in \mathbf{W} : \mu^*(w) \in \mathscr{I} \implies \mu(w) \in \mathscr{I}$.
Proof of Step 1: Consider a worker w for whom $\mu^*(w) \in \mathscr{I}$ and a firm F such that $F \notin \mathscr{I}$. By Lemma 6, (μ^*, s^*) has No Envy, and so it must be the case that $\alpha_w\left(\mu^*(w)\right) + s^*\left(\mu^*(w)\right) \geq \alpha_w(F) + s^*(F)$. $F \notin \mathscr{I}$ implies that $s^*(F) \geq s(F)$, while $\mu^*(w) \in \mathscr{I}$ implies that $s\left(\mu^*(w)\right) > s^*\left(\mu^*(w)\right)$. Combining these inequalities implies that

$$\alpha_w\left(\mu^*(w)\right) + s\left(\mu^*(w)\right) > \alpha_w(F) + s(F).$$

Thus, if $\mu(w) = F$, then (μ, s) would lack No Envy. By Proposition 1, this contradicts the assumption that (μ, s) is stable.

Step 2: $\sum_{F\in\mathscr{I}} \left|\mu^*(F)\right| \le \sum_{F\in\mathscr{I}} \left|\mu(F)\right|$.

Step 2 follows directly from Step 1.

Step 3: $\forall F \in \mathscr{I} : \left|\mu^*(F)\right| \ge \left|\mu(F)\right|$.

Proof of Step 3: Assume towards a contradiction that there exists a firm $F \in \mathscr{I}$ such that $\left|\mu(F)\right| > \left|\mu^*(F)\right|$. By Lemma 6, $s^*(F) \ge \Delta^+_{\mu^*}(F)$. By Lemma 1, y_F has decreasing differences, which with $\left|\mu(F)\right| > \left|\mu^*(F)\right|$ implies that $\Delta^+_{\mu^*}(F) \ge \Delta^-_{\mu}(F)$. Thus $s^*(F) \ge \Delta^-_{\mu}(F)$. Given that $F \in \mathscr{I} : s(F) > s^*(F)$. In summary:

$$s(F) > \Delta^-_{\mu}(F),$$

which means that (μ, s) lacks No Firing. By Proposition 1, this contradicts the assumption that (μ, s) is stable.

Step 4: $\forall F \in \mathscr{I}, \left|\mu^*(F)\right| = \left|\mu(F)\right|$.

Step 4 follows from steps 2 and 3.

Step 5: $\forall w \in \mathbf{W}$: $\mu^*(w) \in \mathscr{I} \iff \mu(w) \in \mathscr{I}$.

Proof of Step 5: By Step 4: $\left|\{w : \mu(w) \in \mathscr{I}\}\right| = \left|\{w : \mu^*(w) \in \mathscr{I}\}\right|$. By Step 1: $\{w : \mu^*(w) \in \mathscr{I}\} \subseteq \{w : \mu(w) \in \mathscr{I}\}$. Thus, $\{w : \mu^*(w) \in \mathscr{I}\} = \{w : \mu(w) \in \mathscr{I}\}$.

Step 6: If for some $G \in \mathbf{F}$ there exists $F \in \mathscr{I}$ and $w \in \mu^*(G)$ such that $\alpha_w(F) + s^*(F) = \alpha_w(G) + s^*(G)$, then $G \in \mathscr{I}$.

Proof of Step 6: Given that $F \in \mathscr{I}$, $\alpha_w(F) + s(F) > \alpha_w(F) + s^*(F)$. By Proposition 1, both (μ, s) and (μ^*, s^*) have No Envy. By No Envy for (μ, s): $\alpha_w\left(\mu(w)\right) + s\left(\mu(w)\right) \ge \alpha_w(F) + s(F)$ and by No Envy for (μ^*, s^*): $\alpha_w(G) + s^*(G) \ge \alpha_w\left(\mu(w)\right) + s^*\left(\mu(w)\right)$.

Combining these inequalities with the equality $\alpha_w(G) + s^*(G) = \alpha_w(F) + s^*(F)$ implies that

$$\alpha_w\left(\mu(w)\right) + s\left(\mu(w)\right) > \alpha_w\left(\mu(w)\right) + s^*\left(\mu(w)\right),$$

and thus $\mu(w) \in \mathscr{I}$. By Step 5 and the fact that $G = \mu^*(w)$: $G \in \mathscr{I}$.

Step 7: $\mathscr{I} = \emptyset$, and thus $s^* \geqq s$.

Proof of Step 7: Assume towards a contradiction that there exists $F_1 \in \mathscr{I}$. By Step 4, $\left|\mu^*(F_1)\right| = \left|\mu(F_1)\right|$, and so $\Delta^-_{\mu^*}(F_1) = \Delta^-_{\mu}(F_1)$. Given that $F_1 \in \mathscr{I}$, $s(F_1) > s^*(F_1)$. Given that (μ, s) is stable, Proposition 1 tells us that it has No Firing, and so $s(F_1) \le \Delta^-_{\mu}(F_1)$. Combining these expressions implies that $s^*(F_1) < \Delta^-_{\mu^*}(F_1)$.

Given that $s^*(F_1) < \Delta^-_{\mu^*}(F_1)$, Lemma 7 tells us that there exists a list of firms $(F_k)_{k=1}^N$ and workers $(w_k)_{k=2}^N$ such that each $w_k \in \mu^*(F_k)$, $s^*(F_N) = \Delta^-_{\mu^*}(F_N)$, and, for each k, $\alpha_{w_k}(F_{k-1}) + s^*(F_{k-1}) = \alpha_{w_k}(F_k) + s^*(F_k)$. By iteratively applying Step 6, this implies that $F_N \in \mathscr{I}$. By Step 4, $\left|\mu^*(F_N)\right| = \left|\mu(F_N)\right|$, and so $\Delta^-_{\mu^*}(F_N) = \Delta^-_{\mu}(F_N)$. So, $s(F_N) > s^*(F_N) = \Delta^-_{\mu^*}(F_N) = \Delta^-_{\mu}(F_N)$. The outcome (μ, s) thus lacks No Firing. By Proposition 1, this contradicts the assumption that it is stable. $\square$

Proposition 3. *Consider an inefficient stable outcome (μ, s). There exists a salary s', a firm F, and a worker w such that $s' < \Delta^+_\mu(F)$, and w strictly prefers to work for firm F at salary s' than for firm $\mu(w)$ at salary $s(\mu(w))$.*

Proof. Consider an inefficient stable outcome (μ, s). We will show that there exists a worker w and a firm $F \neq \mu(w)$ such that

$$\alpha_w(\mu(w)) + s(\mu(w)) < \alpha_w(F) + s'.$$

and that $s' < \Delta^+_\mu(F)$. Combining inequalities, this is equivalent to

$$\alpha_w(\mu(w)) + s(\mu(w)) - \alpha_w(F) < \Delta^+_\mu(F). \tag{14}$$

Let (μ^*, s^*) be the efficient stable outcome constructed in Section 3. By Lemmas 2 and 3 there exists an acyclic replacement chain $\chi = \left((w_k)_{k=0}^{N-1}, (F_k)_{k=0}^{N}\right)$ from μ to μ^* such that value$(\mu + \chi) >$ value(μ). We will show that inequality (14) holds for the replacement chain χ's last firm F_N and its last worker w_{N-1}:

$$\alpha_{w_{N-1}}(\mu(w_{N-1})) + s(\mu(w_{N-1})) - \alpha_{w_{N-1}}(F_N) < \Delta^+_\mu(F_N). \tag{15}$$

Doing so will take four steps.

Step 1: If worker w_{N-1} strictly prefers (μ^*, s^*) to (μ, s), then inequality (15) holds.
Proof of Step 1: Worker w_{N-1}'s strict preference for (μ^*, s^*), in which they are matched to firm F_N, over (μ, s) implies that

$$\alpha_{w_{N-1}}(F_N) + s^*(F_N) > \alpha_{w_{N-1}}(\mu(w_{N-1})) + s(\mu(w_{N-1})).$$

Since (μ^*, s^*) is stable, Proposition 1 tells us that it has No Firing. Thus, $s^*(F_N) \leq \Delta^-_{\mu^*}(F_N)$. Lemma 2 assured us that $|(\mu + \chi)(F_N)| \leq |\mu^*(F_N)|$, while Lemma 1 assures us that y_{F_N} has decreasing differences; in combination these imply that $\Delta^-_{\mu^*}(F_N) \leq \Delta^-_{\mu+\chi}(F_N)$. By the fact that χ is acyclic, $\Delta^-_{\mu+\chi}(F_N) = \Delta^+_\mu(F_N)$. Combining these expressions and rearranging yields inequality (15).

Step 2: If the replacement chain χ contains a worker w_k who strictly prefers (μ^*, s^*) to (μ, s), then the last worker w_{N-1} will strictly prefer (μ^*, s^*) over (μ, s).
Proof of Step 2: Let w_k strictly prefer (μ^*, s^*) (in which they are matched to firm F_{k+1}) over (μ, s) (in which they are matched to firm F_k):

$$\alpha_{w_k}(F_{k+1}) + s^*(F_{k+1}) > \alpha_{w_k}(F_k) + s(F_k).$$

Since (μ, s) is stable, Proposition 1 tells us that it has No Envy, and so

$$\alpha_{w_k}(F_{k+1}) + s(F_{k+1}) \leq \alpha_{w_k}(F_k) + s(F_k).$$

Combining inequalities we have that $s^*(F_{k+1}) > s(F_{k+1})$.

Now consider worker w_{k+1}, who in $\left(\mu, s\right)$ is matched to F_{k+1} and who in $\left(\mu^*, s^*\right)$ is matched to F_{k+2}. Since $\left(\mu^*, s^*\right)$ is stable, Proposition 1 tells us that it has No Envy, and so

$$\alpha_{w_{k+1}}(F_{k+2}) + s^*(F_{k+2}) \geq \alpha_{w_{k+1}}(F_{k+1}) + s^*(F_{k+1}).$$

Given that $s^*(F_{k+1}) > s(F_{k+1})$, this implies that w_{k+1} strictly prefers $\left(\mu^*, s^*\right)$ to $\left(\mu, s\right)$:

$$\alpha_{w_{k+1}}(F_{k+2}) + s^*(F_{k+2}) > \alpha_{w_{k+1}}(F_{k+1}) + s(F_{k+1}).$$

By induction, each worker w_{k+j}, with $j \geq 0$, strictly prefers $\left(\mu^*, s^*\right)$ to $\left(\mu, s\right)$. That includes the last worker w_{N-1}.

Step 3: If the replacement chain χ contains no worker w_k who strictly prefers $\left(\mu^*, s^*\right)$ to $\left(\mu, s\right)$, then inequality (15) holds.

Proof of Step 3: Assume that the replacement chain χ contains no worker w_k who strictly prefers $\left(\mu^*, s^*\right)$ to $\left(\mu, s\right)$. By Theorem 2, this implies that each worker in the replacement chain is indifferent between $\left(\mu^*, s^*\right)$ and $\left(\mu, s\right)$, since the former is preferred by all workers. As such:

$$\sum_{k=0}^{N-1} \left[\alpha_{w_k}(F_{k+1}) - \alpha_{w_k}(F_k) + s^*(F_{k+1}) - s(F_k)\right] = 0$$

Note that

$$\sum_{k=0}^{N-1} \left[s^*(F_{k+1}) - s(F_k)\right] = \sum_{k=1}^{N-1} \left[s^*(F_k) - s(F_k)\right] + s^*(F_N) - s(F_0),$$

and that, by Theorem 2, for each k: $s^*(F_k) \geq s(F_k)$. Thus,

$$\sum_{k=0}^{N-1} \left[\alpha_{w_k}(F_{k+1}) - \alpha_{w_k}(F_k)\right] + s^*(F_N) - s(F_0) \leq 0.$$

Since $\left(\mu, s\right)$ is stable, Proposition 1 tells us that it has No Firing, and so $s(F_0) \leq \Delta_\mu^-(F_0)$. Thus:

$$\sum_{k=0}^{N-1} \left[\alpha_{w_k}(F_{k+1}) - \alpha_{w_k}(F_k)\right] + s^*(F_N) - \Delta_\mu^-(F_0) \leq 0. \tag{16}$$

Given that value$\left(\mu + \chi\right) >$ value$\left(\mu\right)$, it must be the case that

$$\sum_{k=0}^{N-1} \left[\alpha_{w_k}(F_{k+1}) - \alpha_{w_k}(F_k)\right] + \Delta_\mu^+(F_N) - \Delta_\mu^-(F_0) > 0. \tag{17}$$

Inequalities (16) and (17) imply that $\Delta_\mu^+(F_N) > s^*(F_N)$.

Worker w_{N-1} is indifferent between $\left(\mu^*, s^*\right)$, in which she is matched to firm F_N, and $\left(\mu, s\right)$:

$$\alpha_{w_{N-1}}\left(\mu(w_{N-1})\right) + s\left(\mu(w_{N-1})\right) = \alpha_{w_{N-1}}(F_N) + s^*(F_N).$$

With $\Delta^+_\mu(F_N) > s^*(F_N)$, this implies inequality (15).

Step 4: Inequality (15) holds.

Proof of Step 4: By Step 2, if any worker in the replacement chain χ strictly prefers (μ^*, s^*) to (μ, s), then w_{N-1} will strictly prefer (μ^*, s^*) to (μ, s). By Step 1, if w_{N-1} strictly prefers (μ^*, s^*) to (μ, s), then inequality (15) holds.

On the other hand, if no worker in the replacement chain χ strictly prefers (μ^*, s^*) to (μ, s), then Step 3 tells us that inequality (15) holds. Thus, regardless of whether a worker in the replacement chain χ strictly prefers (μ^*, s^*) to (μ, s), inequality (15) must hold.

□

Theorem 3.

1. *When firms' production functions are public information, a strategyproof mechanism can elicit workers' preferences to implement an efficient stable outcome.*
2. *When firms' production functions are private information, no strategyproof mechanism can generally implement an efficient stable outcome.*

Proof. The second claim of Theorem 3 is proved by Example 2. The proof of the first claim of Theorem 3 is as follows.

Consider the mechanism which asks each worker for her amenities and then implements the efficient stable outcome constructed in Section 3. Note that regardless of the veracity of workers' reported amenities, the mechanism will use the true production functions since we assumed they are public information. We will show that, under such a mechanism, it is a weakly-dominant strategy for each worker to report her true amenities.

Let $\alpha_w \equiv (\alpha_w(F))_{F\in\mathbf{F}}$ concatenate each worker w's amenities. Let α°_w represent worker w's reported amenities. Fix a particular worker $\hat{w} \in \mathbf{W}$. Assume towards a contradiction that there exists a report $\alpha^\circ_{\hat{w}} \neq \alpha_{\hat{w}}$ such that $\hat{w}$ strictly benefits from reporting $\alpha^\circ_{\hat{w}}$, given the other workers' reports. Let (μ, s) denote the outcome produced when all workers w (including $w = \hat{w}$) report α°_w. Let (μ^*, s^*) denote the outcome produced when all workers $w \neq \hat{w}$ report α°_w, while worker $\hat{w}$ truthfully reports $\alpha_{\hat{w}}$. Let $F^* \equiv \mu^*(\hat{w})$ and let $F^\circ \equiv \mu(\hat{w})$. Our assumption that $\hat{w}$ benefits from misreporting requires:

$$\alpha_{\hat{w}}(F^*) + s^*(F^*) < \alpha_{\hat{w}}(F^\circ) + s(F^\circ). \tag{18}$$

By Lemma 6, both (μ, s) and (μ^*, s^*) have No Envy for their respective reports, though not necessarily for the true amenities. For clarity, we will say an outcome has No Reported Envy if that outcome would have No Envy if reported amenities were true.

Since (μ^*, s^*) has No Reported Envy:

$$\alpha_{\hat{w}}(F^*) + s^*(F^*) \geq \alpha_{\hat{w}}(F^\circ) + s^*(F^\circ)$$

with inequality (18), this implies that $s^*(F^\circ) < s(F^\circ)$. Let $\mathscr{I} = \{F : s(F) > s^*(F)\}$. We have shown that $F^\circ \in \mathscr{I}$. We will prove the contradiction that $\mathscr{I} = \emptyset$. The proof from this point is similar to that for Theorem 2. The only difference is that we will here require No Reported Envy rather than No Envy.

Step 1: $\forall w \in \mathbf{W} : \mu^*(w) \in \mathscr{I} \implies \mu(w) \in \mathscr{I}$.

Proof of Step 1: We showed above that $F^\circ = \mu(\hat{w}) \in \mathscr{I}$, and thus it remains to show that the claim is true for all $w \neq \hat{w}$. Consider a worker $w \neq \hat{w}$ for whom $\mu^*(w) \in \mathscr{I}$ and a firm F such that $F \notin \mathscr{I}$. Since (μ^*, s^*) has No Reported Envy, it must be the case that $\alpha^\circ_w(\mu^*(w)) + s^*(\mu^*(w)) \geq \alpha^\circ_w(F) + s^*(F)$. $F \notin \mathscr{I}$ implies that $s^*(F) \geq s(F)$, while $\mu^*(w) \in \mathscr{I}$ implies that $s(\mu^*(w)) > s^*(\mu^*(w))$. Combining these inequalities implies that

$$\alpha^\circ_w(\mu^*(w)) + s(\mu^*(w)) > \alpha^\circ_w(F) + s(F).$$

Thus, $\mu(w) = F$ would imply (μ, s) lacks No Reported Envy – a contradiction.

Step 2: $\sum_{F \in \mathscr{I}} |\mu^*(F)| \leq \sum_{F \in \mathscr{I}} |\mu(F)|$.

Step 2 follows directly from Step 1.

Step 3: $\forall F \in \mathscr{I} : |\mu^*(F)| \geq |\mu(F)|$.

The proof of Step 3 is identical to the proof of Step 3 of Theorem 2.

Step 4: $\forall F \in \mathscr{I}, |\mu^*(F)| = |\mu(F)|$.

Step 4 follows from Steps 2 and 3.

Step 5: $\forall w \in \mathbf{W}$: $\mu^*(w) \in \mathscr{I} \iff \mu(w) \in \mathscr{I}$.

The proof of Step 5 is identical to the proof of Step 5 of Theorem 2.

Step 6: If for some $G \in \mathbf{F}$, there exists $F \in \mathscr{I}$ and $w \in \mu^*(G)$ such that $\alpha^\circ_w(F) + s^*(F) = \alpha^\circ_w(G) + s^*(G)$, then $G \in \mathscr{I}$.

Proof of Step 6: We showed above that $F^\circ = \mu(\hat{w}) \in \mathscr{I}$, and thus it remains to show that the claim is true for all $w \neq \hat{w}$. Given that $F \in \mathscr{I}$, $\alpha^\circ_w(F) + s(F) > \alpha^\circ_w(F) + s^*(F)$. By No Reported Envy for (μ, s): $\alpha^\circ_w(\mu(w)) + s(\mu(w)) \geq \alpha^\circ_w(F) + s(F)$ and by No Reported Envy for (μ^*, s^*): $\alpha^\circ_w(G) + s^*(G) \geq \alpha^\circ_w(\mu(w)) + s^*(\mu(w))$.

Combining these inequalities with the equality $\alpha^\circ_w(G) + s^*(G) = \alpha^\circ_w(F) + s^*(F)$ implies that

$$\alpha^\circ_w(\mu(w)) + s(\mu(w)) > \alpha^\circ_w(\mu(w)) + s^*(\mu(w)).$$

and thus $\mu(w) \in \mathscr{I}$. By Step 5 and the fact that $G = \mu^*(w)$: $G \in \mathscr{I}$.

Step 7: $\mathscr{I} = \emptyset$.

The proof of Step 7 is identical to the proof of Step 7 of Theorem 2.

Step 7 contradicts our earlier result that $F^\circ \in \mathscr{I}$, completing the proof. □

Online Appendix to "Job Matching without Price Discrimination"

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July 15, 2026

B Proofs of auxiliary results

In this appendix, we present proofs of all our lemmas.

Lemma 1. *Every firm's production function has decreasing differences.*

Proof. Assume towards a contradiction that some firm F's production function does not have decreasing differences:

$$\exists N \in \mathbb{N} \text{ such that } y_F(N) - y_F(N-1) > y_F(N-1) - y_F(N-2). \tag{19}$$

Without loss of generality, let N be the smallest integer such that inequality (19) holds for firm F.

Consider the salary $s_\epsilon \equiv y_F(N-1) - y_F(N-2) + \epsilon$, where $\epsilon \geq 0$. Given that $y_F(0) = 0$, firm F's profit from employing N workers at salary s_ϵ is

$$\pi_F(N, s_\epsilon) = \sum_{i=1}^{N} \left[y_F(i) - y_F(i-1) - s_\epsilon \right].$$

This implies that when $\epsilon = 0$, the marginal profit from hiring the ith worker is

$$\begin{aligned} & y_F(i) - y_F(i-1) - s_0 \\ = \; & y_F(i) - y_F(i-1) - \big(y_F(N-1) - y_F(N-2) \big). \end{aligned}$$

By the assumption that N is the smallest integer such that inequality (19) holds, the firm's marginal profit will be weakly positive for $i < N$ and strictly positive for $i = N$. Thus:

$$\forall M < N : \pi_F(N, s_0) > \pi_F(M, s_0).$$

Moreover, the continuity of the profit function with respect to the salary implies that the inequality will continue to hold for all ϵ sufficiently close to 0:

$$\exists \epsilon > 0 : \forall M < N : \pi_F(N, s_\epsilon) > \pi_F(M, s_\epsilon). \tag{20}$$

However, for any $\epsilon > 0$, the marginal profit from hiring the $(N-1)$th worker is negative. Thus:

$$\forall \epsilon > 0 : \pi_F(N-1, s_\epsilon) < \pi_F(N-2, s_\epsilon). \tag{21}$$

In combination, expressions (20) and (21) contradict the gross substitutes assumption. □

Lemma 2. *Let μ and μ^* be matchings such that value$(\mu^*) >$ value(μ). There exists a replacement chain χ from μ to μ^* such that value$(\mu+\chi) >$ value(μ). Moreover, for each firm F: $|(\mu+\chi)(F)| \leq \max\{|\mu(F)|, |\mu^*(F)|\}$.*

Proof. Let us first define a maximal replacement chain. $\chi = \left((w_k)_{k=0}^{N-1}, (F_k)_{k=0}^{N}\right)$ is a **maximal** chain from μ to μ' if it cannot be extended in either direction: $\mu'(F_0) \subseteq (\mu+\chi)(F_0)$ and $(\mu+\chi)(F_N) \subseteq \mu'(F_N)$.

Moreover, for a matching μ', if χ is such that

$$\forall k \in 0, \ldots, N-1 : w_k \in \mu'(F_{k+1}),$$

then we define $\mu' - \chi$ to be the matching constructed by moving each worker w_k from F_{k+1} to F_k:

$$(\mu' - \chi)(w) = \begin{cases} F_k & \text{if } w = w_k; \\ \mu'(w) & \text{if } w \notin (w_k)_{k=0}^{N-1}. \end{cases}$$

Our proof of Lemma 2 is algorithmic. The state of an algorithm is a matching μ°. The algorithm is as follows:

1. Initialize the algorithm with $\mu^\circ \leftarrow \mu^*$. Then go to 2.
2. If there exists a cyclic replacement chain from μ to μ°, go to 3. Otherwise, go to 4.
3. Let χ be a cyclic replacement chain from μ to μ°. If value$(\mu+\chi) >$ value(μ), then χ is the required replacement chain, and the algorithm can terminate. If not, set $\mu^\circ \leftarrow \mu^\circ - \chi$, and go to 2.
4. If there exists a maximal acyclic replacement chain from μ to μ°, go to 5. Otherwise, go to 6.
5. Let χ be a maximal acyclic replacement chain from μ to μ°. If value$(\mu+\chi) >$ value(μ), then χ is the required replacement chain, and the algorithm can terminate. If not, set $\mu^\circ \leftarrow \mu^\circ - \chi$, and go to 2.
6. Terminate the algorithm.

When the algorithm does not terminate on lines 3 or 5, the state μ° becomes more similar to μ. When $\mu^\circ = \mu$, there is no replacement chain from μ to μ°, and so the algorithm will terminate at line 6. Because the matching is discrete, this means that the algorithm must terminate eventually.

Lemma 2 will hold provided that the algorithm never terminates at line 6, and that for each replacement chain χ proposed in lines 3 and 5: χ is a replacement chain from μ to μ^*, and for each firm F: $\left|(\mu+\chi)(F)\right| \leq \max\left\{\left|\mu(F)\right|, \left|\mu^*(F)\right|\right\}$. We prove these results in the following 6 steps.

Step 1: If an acyclic replacement chain $\chi = \left((w_k)_{k=0}^{N-1}, (F_k)_{k=0}^{N}\right)$ is maximal from μ to μ' then $\left|\mu(F_N)\right| < \left|\mu'(F_N)\right|$ and $\left|\mu'(F_0)\right| < \left|\mu(F_0)\right|$.

Proof of Step 1: Given that χ is maximal, $\left(\mu+\chi\right)(F_N) \subseteq \mu'(F_N)$. Thus $\left|\left(\mu+\chi\right)(F_N)\right| \leq \left|\mu'(F_N)\right|$. Given that χ is acyclic, $\left|\left(\mu+\chi\right)(F_N)\right| = \left|\mu(F_N)\right|+1$. Combining inequalities we have that $\left|\mu(F_N)\right| < \left|\mu'(F_N)\right|$. The proof that $\left|\mu'(F_0)\right| < \left|\mu(F_0)\right|$ is similar.

Step 2: Given two matchings $\mu \neq \mu'$, either there either exists a cyclic replacement chain from μ to μ' or there exists a maximal acyclic replacement chain from μ to μ'.

Proof of Step 2: Given that $\mu \neq \mu'$ there must exist a worker w_0 with $\mu(w_0) \neq \mu'(w_0)$. Let χ be the trivial replacement chain $\left((w_0), \left(\mu(w_0), \mu'(w_0)\right)\right)$. Iteratively extend χ by adding additional workers. Given the finite number of firms, either χ will eventually be a cyclic replacement chain or it will eventually be a maximal acyclic replacement chain.

Step 3: value$\left(\mu^\circ\right)$ is weakly increasing as the algorithm proceeds.

Proof of Step 3: We first show that value$\left(\mu^\circ\right)$ is weakly increasing as the algorithm proceeds. The state matching μ° is altered in lines 3 and 5. In line 3, the replacement chain $\chi = \left((w_k)_{k=0}^{N-1}, (F_k)_{k=0}^{N}\right)$ is cyclic, and thus it does not change the number of workers matched to any firm. Thus:

$$\text{value}\left(\mu^\circ\right) - \text{value}\left(\mu^\circ - \chi\right) = \sum_{k=0}^{N-1}\left[\alpha_{w_k}(F_{k+1}) - \alpha_{w_k}(F_k)\right] = \text{value}\left(\mu+\chi\right) - \text{value}\left(\mu\right),$$

which is non-positive if the algorithm does not terminate. Thus, value$\left(\mu^\circ\right) \leq$ value$\left(\mu^\circ - \chi\right)$.

In line 5, the replacement chain $\chi = \left((w_k)_{k=0}^{N-1}, (F_k)_{k=0}^{N}\right)$ is acyclic. It thus removes a worker from F_0 and adds a worker to F_N. As such:

$$\text{value}\left(\mu+\chi\right) - \text{value}\left(\mu\right) = \Delta_\mu^+(F_N) - \Delta_\mu^-(F_0) + \sum_{k=0}^{N-1}\left[\alpha_{w_k}(F_{k+1}) - \alpha_{w_k}(F_k)\right]; \tag{22}$$

$$\text{value}\left(\mu^\circ\right) - \text{value}\left(\mu^\circ - \chi\right) = \Delta_{\mu^\circ}^-(F_N) - \Delta_{\mu^\circ}^+(F_0) + \sum_{k=0}^{N-1}\left[\alpha_{w_k}(F_{k+1}) - \alpha_{w_k}(F_k)\right]. \tag{23}$$

The replacement chain χ proposed in line 5 is also maximal from μ to μ°. By Step 1, $\left|\mu(F_N)\right| < \left|\mu^\circ(F_N)\right|$. Thus:

$$\begin{aligned}\Delta_{\mu^\circ}^-(F_N) &\equiv y_{F_N}\left(\left|\mu^\circ(F_N)\right|\right) - y_{F_N}\left(\left|\mu^\circ(F_N)\right| - 1\right)\\ &\leq y_{F_N}\left(\left|\mu(F_N)\right| + 1\right) - y_{F_N}\left(\left|\mu(F_N)\right|\right) \equiv \Delta_\mu^+(F_N),\end{aligned}$$

where the second line follows from $|\mu(F_N)| < |\mu^\circ(F_N)|$ and the fact that, by Lemma 1, y_{F_N} has decreasing differences. Similarly, $\Delta^+_{\mu^\circ}(F_0) \geq \Delta^-_{\mu}(F_0)$. Combining these inequalities with equations (22) and (23) yields

$$\text{value}\left(\mu^\circ\right) - \text{value}\left(\mu^\circ - \chi\right) \leq \text{value}\left(\mu + \chi\right) - \text{value}\left(\mu\right).$$

If the algorithm does not terminate then value$\left(\mu + \chi\right)$−value$\left(\mu\right) \leq 0$. Thus, value$\left(\mu^\circ\right) \leq$ value$\left(\mu^\circ - \chi\right)$. This completes the proof that value$\left(\mu^\circ\right)$ is weakly increasing as the algorithm proceeds.

Step 4: The algorithm never terminates at line 6.

Proof of Step 4: Initially, value$\left(\mu^\circ\right)$ = value$\left(\mu^*\right)$ > value$\left(\mu\right)$. With Step 3, this implies that at every stage of the algorithm:

$$\text{value}\left(\mu^\circ\right) > \text{value}\left(\mu\right) \tag{24}$$

But the algorithm only reaches line 6 if there is neither a cyclic replacement chain from μ to μ° nor a maximal acyclic replacement chain from μ to μ°. By Step 2, this would require that $\mu^\circ = \mu$ and thus value$\left(\mu^\circ\right)$ = value$\left(\mu\right)$. This would contradict Inequality (24).

Step 5: Each replacement chain proposed in lines 3 and 5 is a replacement chain from μ to μ^*.

Proof of Step 5: Let $\chi = \left((w_k)_{k=0}^{N-1}, (F_k)_{k=0}^{N}\right)$ be a candidate replacement chain proposed in lines 3 or 5. We must show that $\forall k: w_k \in \mu(F_k) \cap \mu^*(F_{k+1})$. Given that χ is a replacement chain from μ to μ°: $w_k \in \mu(F_k) \cap \mu^\circ(F_{k+1})$. Given that $F_k \neq F_{k+1}$, that $F_k = \mu(w_k)$, and that $F_{k+1} = \mu^\circ(w_k)$, this implies that $\mu^\circ(w_k) \neq \mu(w_k)$. But when the state μ° is updated in lines 3 and 5, workers are only ever moved from their match in μ^* to their match in μ. Thus, given that $\mu^\circ(w_k) \neq \mu(w_k)$ it must be the case that $\mu^\circ(w_k) = \mu^*(w_k)$. Given that $F_{k+1} = \mu^\circ(w_k)$, this completes the proof that, for each k,

$$w_k \in \mu(F_k) \cap \mu^*(F_{k+1}).$$

Step 6: As the algorithm runs, the state matching μ° is such that $\forall F: |\mu^\circ(F)| \leq \max\left\{|\mu(F)|, |\mu^*(F)|\right\}$.

Proof of Step 6: Assume towards a contradiction that, at some point of the algorithm, this is not the case. Given that μ° is initially set equal to μ^*, this requires that there be a point in the algorithm such that

$$\forall F: |\mu^\circ(F)| \leq \max\left\{|\mu(F)|, |\mu^*(F)|\right\}; \quad \exists F_0: \left|\left(\mu^\circ - \chi\right)(F_0)\right| > \max\left\{|\mu(F_0)|, |\mu^*(F_0)|\right\}, \tag{25}$$

where $\chi = \left((w_k)_{k=0}^{N-1}, (F_k)_{k=0}^{N}\right)$ is a replacement chain proposed in lines 3 or 5. Replacement chains proposed in line 3 are cyclic and thus do not change the number of workers employed at any firm. Thus, it must be the case that χ is proposed in line 5.

Replacement chains proposed in line 5 are maximal from μ to μ°, and so, by Step 1, $|\mu^\circ(F_0)| < |\mu(F_0)|$. Subtracting the replacement chain χ moves at most one worker to firm F_0, and so $\left|\left(\mu^\circ - \chi\right)(F_0)\right| \leq |\mu(F_0)| \leq \max\left\{|\mu(F_0)|, |\mu^*(F_0)|\right\}$, which contradicts expression (25).

Step 7: Each replacement chain χ proposed in lines 3 and 5 is such that for each firm F: $\left|(\mu+\chi)(F)\right| \leq \max\{\left|\mu(F)\right|, \left|\mu^*(F)\right|\}$.

Proof of Step 7: This proof is similar to that of Step 6. Assume towards a contradiction that some replacement chain $\chi = \left((w_k)_{k=0}^{N-1}, (F_k)_{k=0}^{N}\right)$ proposed in line 3 or 5 is such that

$$\left|(\mu+\chi)(F_N)\right| > \max\{\left|\mu(F_N)\right|, \left|\mu^*(F_N)\right|\}. \tag{26}$$

Replacement chains proposed in line 3 are cyclic and thus do not change the number of workers employed at any firm. Thus, it must be the case that χ is proposed in line 5.

Replacement chains proposed in line 5 are maximal from μ to μ°, and so, by Step 1, $\left|\mu(F_N)\right| < \left|\mu^\circ(F_N)\right|$. The replacement chain χ moves at most one worker to firm F_N, and so $\left|(\mu+\chi)(F_N)\right| \leq \left|\mu^\circ(F_N)\right|$. By Step 6, $\left|\mu^\circ(F_N)\right| \leq \max\{\left|\mu(F_N)\right|, \left|\mu^*(F_N)\right|\}$. Combining inequalities, $\left|(\mu+\chi)(F_N)\right| \leq \max\{\left|\mu(F_N)\right|, \left|\mu^*(F_N)\right|\}$. This contradicts expression (26).

□

Lemma 3. *Let (μ, s) be a stable outcome. There exists no cyclic replacement chain χ such that value$(\mu+\chi) >$ value(μ).*

Proof. Assume towards a contradiction that there exists a cyclic replacement chain $\chi = \left((w_k)_{k=0}^{N-1}, (F_k)_{k=0}^{N}\right)$ such that value$(\mu+\chi) >$ value(μ). Given that χ is cyclic, it does not change the number of workers employed by any firm. Thus, the only difference between value$(\mu+\chi)$ and value(μ) is workers' amenities. It thus follows from value$(\mu+\chi) >$ value(μ) that

$$\sum_{k=0}^{N-1} \left[\alpha_{w_k}(F_{k+1}) - \alpha_{w_k}(F_k)\right] > 0. \tag{27}$$

Given that χ is cyclic, $F_N = F_0$, and so inequality (27) implies that

$$\sum_{k=0}^{N-1} \left[\alpha_{w_k}(F_{k+1}) - \alpha_{w_k}(F_k) + s(F_{k+1}) - s(F_k)\right] > 0.$$

As such, there must be some k such that $\alpha_{w_k}(F_{k+1}) - \alpha_{w_k}(F_k) + s(F_{k+1}) - s(F_k) > 0$. Given that $\mu(w_k) = F_k$, this implies that

$$\alpha_{w_k}(F_{k+1}) + s(F_{k+1}) > \alpha_{w_k}\left(\mu(w_k)\right) + s\left(\mu(w_k)\right).$$

Thus, (μ, s) lacks No Envy. By Proposition 1 it cannot be stable. □

Lemma 4. *An outcome with Marginal Product Salaries has both No Firing and No Poaching.*

Proof. That an outcome with Marginal Product Salaries has No Firing follows from $s(F) \leq \Delta_\mu^-(F)$.

Fix a firm F and a salary $s' > s(F)$. Because $s(F) \geq \Delta^+_\mu(F)$:

$$s' > \Delta^+_\mu(F) \equiv y_F\left(\left|\mu(F)\right| + 1\right) - y_F\left(\left|\mu(F)\right|\right). \tag{28}$$

To show that the outcome has No Poaching, we must show that there exists no employment level L such that $\left|\mu(F)\right| < L \leq L_F(s', s)$ and that $\pi_F(L, s') \geq \pi_F\left(\left|\mu(F)\right|, s(F)\right)$. Consider any employment level $L > \left|\mu(F)\right|$:

$$\begin{aligned}\pi_F\left(L, s'\right) = \sum_{i=1}^{L}\left[y_F(i) - y_F(i-1) - s'\right] &= \sum_{i=1}^{\left|\mu(F)\right|}\left[y_F(i) - y_F(i-1) - s'\right] + \sum_{i=\left|\mu(F)\right|+1}^{L}\left[y_F(i) - y_F(i-1) - s'\right] \\ &< \sum_{i=1}^{\left|\mu(F)\right|}\left[y_F(i) - y_F(i-1) - s(F)\right] + \sum_{i=\left|\mu(F)\right|+1}^{L}\left[y_F(i) - y_F(i-1) - s'\right],\end{aligned} \tag{29}$$

where the inequality follows from $s(F) < s'$. By Lemma 1, each firm's production function has decreasing differences, and so

$$\sum_{i=\left|\mu(F)\right|+1}^{L}\left[y_F(i) - y_F(i-1) - s'\right] \leq \sum_{i=\left|\mu(F)\right|+1}^{L}\left[y_F\left(\left|\mu(F)\right| + 1\right) - y_F\left(\left|\mu(F)\right|\right) - s'\right],$$

which, in combination with expression (28), implies that

$$\sum_{i=\left|\mu(F)\right|+1}^{L}\left[y_F(i) - y_F(i-1) - s'\right] < 0.$$

With inequality (29), this implies that

$$\pi_F\left(L, s'\right) < \sum_{i=1}^{\left|\mu(F)\right|}\left[y_F(i) - y_F(i-1) - s(F)\right] = \pi_F\left(\left|\mu(F)\right|, s(F)\right).$$

□

Lemma 5. *Let μ be hedonic efficient. For any firms $(F_k)_{k=1}^N$ with $F_1 = F_N$: $\sum_{k=1}^{N-1} d^1_\mu(F_k, F_{k+1}) \leq 0$.*

Proof. Assume towards a contradiction that there exists a list of firms $(F_k)_{k=1}^N$ such that $F_1 = F_N$ and that $\sum_{k=1}^{N-1} d^1_\mu(F_k, F_{k+1}) > 0$. Given that $d^1_\mu(F_k, F_{k+1})$ would equal $-\infty$ if F_{k+1} were unmatched it must be the case that each F_{k+1} is matched to at least one worker. For each $k = 1, 2, ..., N-1$, let $w_k \in \arg\max_{w \in \mu(F_{k+1})} \{\alpha_w(F_k) - \alpha_w(F_{k+1})\}$. By the definition of d^1_μ it follows that $d^1_\mu(F_k, F_{k+1}) = \alpha_{w_k}(F_k) - \alpha_{w_k}(F_{k+1})$. Substituting this into the above inequality yields

$$\sum_{k=1}^{N-1}\left[\alpha_{w_k}(F_k) - \alpha_{w_k}(F_{k+1})\right] > 0. \tag{30}$$

Define the replacement chain $\chi = \left((w_k)_{k=1}^{N-1}, (F_k)_{k=1}^{N}\right)$. Recall that each $w_k \in \mu(F_{k+1})$ and thus the matching $\mu - \chi$ is the same as μ except with each worker w_k moved from firm F_{k+1} to firm F_k. Given that $F_N = F_1$, each firm has the same number of workers in $\mu - \chi$ as in μ, and so:

$$\text{value}\left(\mu - \chi\right) - \text{value}\left(\mu\right) = \sum_{k=1}^{N-1} \left[\alpha_{w_k}(F_k) - \alpha_{w_k}(F_{k+1})\right].$$

With inequality (30), this means that $\text{value}\left(\mu - \chi\right) > \text{value}\left(\mu\right)$. This contradicts the hedonic efficiency of μ. $\square$

Lemma 6. *The outcome* $\left(\mu^*, s^*\right)$ *has No Envy and Marginal Product Salaries.*

Proof. We will prove Lemma 6 in three steps.

Step 1: The outcome $\left(\mu^*, s^*\right)$ has No Envy.

Proof of Step 1: Consider firms F, H and worker $w \in \mu^*(F)$. By definition,

$$s^*(F) = \min_{G \in \mathbf{F} \cup \{\emptyset\}} \left\{\Delta^-_{\mu^*}(G) - d^\infty_{\mu^*}(F, G)\right\}.$$

Thus, there is some firm G' such that $s^*(F) = \Delta^-_{\mu^*}(G') - d^\infty_{\mu^*}(F, G')$. Similarly, H's salary is given by

$$\begin{aligned} s^*(H) \equiv \min_{G \in \mathbf{F} \cup \{\emptyset\}} \left\{\Delta^-_{\mu^*}(G) - d^\infty_{\mu^*}(H, G)\right\} &\leq \Delta^-_{\mu^*}(G') - d^\infty_{\mu^*}(H, G') \\ &\leq \Delta^-_{\mu^*}(G') - \left(d^1_{\mu^*}(H, F) + d^\infty_{\mu^*}(F, G')\right) \\ &= s^*(F) - d^1_{\mu^*}(H, F) \\ &= s^*(F) - \max_{w' \in \mu^*(F)} \{\alpha_{w'}(H) - \alpha_{w'}(F)\} \\ &\leq s^*(F) - \alpha_w(H) + \alpha_w(F). \end{aligned}$$

Thus, $s^*(H) + \alpha_w(H) \leq s^*(F) + \alpha_w(F)$, as No Envy requires.

Step 2: For each firm F, $s^*(F) \leq \Delta^-_{\mu^*}(F)$.

Proof of Step 2: Consider any firm F. By Lemma 5: $d^\infty_{\mu^*}(F, F) = d^1_{\mu^*}(F, F) = 0$. Thus, $s^*(F) \leq \Delta^-_{\mu^*}(F) - d^\infty_{\mu^*}(F, F) = \Delta^-_{\mu^*}(F)$.

Step 3: For each firm F, $s^*(F) \geq \Delta^+_{\mu^*}(F)$.

Proof of Step 3: Assume towards a contradiction that, for some firm F_1: $s^*(F_1) < \Delta^+_{\mu^*}(F_1)$. By the definition of the salary schedule s^*, there exists a list of firms $(F_k)_{k=1}^N$ such that $s^*(F_1) = \Delta^-_{\mu^*}(F_N) - \sum_{k=1}^{N-1} d^1_{\mu^*}(F_k, F_{k+1})$. Combining expressions, we have that

$$\Delta^-_{\mu^*}(F_N) - \sum_{k=1}^{N-1} d^1_{\mu^*}(F_k, F_{k+1}) < \Delta^+_{\mu^*}(F_1).$$

Given that $d^1_{\mu}(F_k, F_{k+1})$ would equal $-\infty$ if F_{k+1} were unmatched it must be the case that each $\mu^*(F_{k+1}) \neq \emptyset$. For each $k = 1, 2, ..., N-1$, let $w_k \in \arg\max_{w \in \mu^*(F_{k+1})} \{\alpha_w(F_k) - \alpha_w(F_{k+1})\}$. By the definition of d^1_μ it follows that $d^1_{\mu^*}(F_k, F_{k+1}) = \alpha_{w_k}(F_k) - \alpha_{w_k}(F_{k+1})$. Substituting this into the above inequality and rearranging implies that

$$\Delta^+_{\mu^*}(F_1) - \Delta^-_{\mu^*}(F_N) + \sum_{k=1}^{N-1} \left[\alpha_{w_k}(F_k) - \alpha_{w_k}(F_{k+1})\right] > 0. \tag{31}$$

Define the replacement chain $\chi = \left((w_k)_{k=1}^{N-1}, (F_k)_{k=1}^{N}\right)$. By inequality (31), the value of μ^* could be improved by moving each worker w_k from firm F_{k+1} to firm F_k: value$\left(\mu^* - \chi\right) >$ value$\left(\mu^*\right)$. This contradicts the efficiency of μ^*, completing the proof of Step 3. □

Lemma 7. *Let $\left(\mu^*, s^*\right)$ be the outcome constructed in Section 3. For each firm F_1, either $s^*(F_1) = \Delta^-_{\mu^*}(F_1)$, or there exists a list of firms $(F_k)_{k=2}^N$ each containing some worker $w_k \in \mu^*(F_k)$, such that each worker w_k is indifferent between firm F_k and firm F_{k-1}:*

$$\alpha_{w_k}(F_{k-1}) + s^*(F_{k-1}) = \alpha_{w_k}(F_k) + s^*(F_k),$$

and $s^*(F_N) = \Delta^-_{\mu^*}(F_N)$.

Proof. Consider some firm F_1. By equation (1), there exists a firm F_N such that

$$s^*(F_1) = \Delta^-_{\mu^*}(F_N) - d^\infty_{\mu^*}(F_1, F_N).$$

If $F_1 = F_N$, then, by Lemma 5, $d^\infty_{\mu^*}(F_1, F_N) = d^1_{\mu^*}(F_1, F_N) = 0$ and so $s^*(F_1) = \Delta^-_{\mu^*}(F_1)$. If $F_1 \neq F_N$, there exists a path of firms $(F_h)_{h=2}^N$ such that

$$s^*(F_1) = \Delta^-_{\mu^*}(F_N) - \sum_{h=1}^{N-1} d^1_\mu(F_h, F_{h+1}).$$

For each $h = 2, \ldots, N$, let $w_h \in \arg\max_{w \in \mu(F_h)} \{\alpha_w(F_{h-1}) - \alpha_w(F_h)\}$. By the definition of d^1_μ it follows that

$$s^*(F_1) = \Delta^-_{\mu^*}(F_N) - \sum_{h=1}^{N-1} \left[\alpha_{w_{h+1}}(F_h) - \alpha_{w_{h+1}}(F_{h+1})\right]. \tag{32}$$

We will show that, for each j,

$$s^*(F_j) = \Delta^-_{\mu^*}(F_N) - \sum_{h=j}^{N-1} \left[\alpha_{w_{h+1}}(F_h) - \alpha_{w_{h+1}}(F_{h+1})\right]. \tag{33}$$

Let us first show that $s^*(F_j) \le \Delta^-_{\mu^*}(F_N) - \sum_{h=j}^{N-1} \left[\alpha_{w_{h+1}}(F_h) - \alpha_{w_{h+1}}(F_{h+1})\right]$. By Lemma 6, $\left(\mu^*, s^*\right)$ has No Envy. No Envy implies that, for each h, $s^*(F_{h+1}) + \alpha_{w_{h+1}}(F_{h+1}) - s^*(F_h) - \alpha_{w_{h+1}}(F_h) \ge 0$. Summing across h yields

$$\sum_{h=j}^{N-1} \left[s^*(F_{h+1}) + \alpha_{w_{h+1}}(F_{h+1}) - s^*(F_h) - \alpha_{w_{h+1}}(F_h)\right] \ge 0.$$

Simplifying the telescoping sum and rearranging yields

$$s^*(F_j) \leq s^*(F_N) - \sum_{h=j}^{N-1} \left[\alpha_{w_{h+1}}(F_h) - \alpha_{w_{h+1}}(F_{h+1})\right].$$

By Lemma 6, the outcome (μ^*, s^*) has Marginal Product Salaries and thus $s^*(F_N) \leq \Delta^-_{\mu^*}(F_N)$. This completes the proof that $s^*(F_j) \leq \Delta^-_{\mu^*}(F_N) - \sum_{h=j}^{N-1}\left[\alpha_{w_{h+1}}(F_h) - \alpha_{w_{h+1}}(F_{h+1})\right]$.

Thus, it remains to show that, for each j, $s^*(F_j) \geq \Delta^-_{\mu^*}(F_N) - \sum_{h=j}^{N-1}\left[\alpha_{w_{h+1}}(F_h) - \alpha_{w_{h+1}}(F_{h+1})\right]$. As before, consider the No Envy requirement that $s^*(F_{h+1}) + \alpha_{w_{h+1}}(F_{h+1}) - s^*(F_h) - \alpha_{w_{h+1}}(F_h) \geq 0$, and sum across h:

$$\sum_{h=1}^{j-1} \left[s^*(F_{h+1}) + \alpha_{w_{h+1}}(F_{h+1}) - s^*(F_h) - \alpha_{w_{h+1}}(F_h)\right] \geq 0.$$

Simplifying and rearranging this inequality yields

$$s^*(F_1) \leq s^*(F_j) - \sum_{h=1}^{j-1} \left[\alpha_{w_{h+1}}(F_h) - \alpha_{w_{h+1}}(F_{h+1})\right].$$

Subtracting equation (32) from both sides yields

$$0 \leq s^*(F_j) - \sum_{h=1}^{j-1} \left[\alpha_{w_{h+1}}(F_h) - \alpha_{w_{h+1}}(F_{h+1})\right] - \Delta^-_{\mu^*}(F_N) + \sum_{h=1}^{N-1} \left[\alpha_{w_{h+1}}(F_h) - \alpha_{w_{h+1}}(F_{h+1})\right],$$

which can be rearranged to find that $s^*(F_j) \geq \Delta^-_{\mu^*}(F_N) - \sum_{h=j}^{N-1}\left[\alpha_{w_{h+1}}(F_h) - \alpha_{w_{h+1}}(F_{h+1})\right]$. This completes the proof of equation (33). The claim that $\alpha_{w_k}(F_{k-1}) + s^*(F_{k-1}) = \alpha_{w_k}(F_k) + s^*(F_k)$ follows from subtracting equation (33) for $j = k$ and $j = k-1$. □

C Relationships Between Our Model and Others

In this appendix, we contrast results and assumptions made by our model to results and assumptions made in the literature.

C.1 A worker can prefer a stable outcome preferred by all firms

Payoffs in matching models frequently form bounded lattices with payoffs on one side of the market dual to payoffs on the other side (Knuth, 1976; Shapley & Shubik, 1971; Hatfield & Milgrom, 2005; Blair, 1988). We will now show that this duality fails in our model: a worker and all firms can all benefit from the shift from one stable outcome to another.

Example C.1 (a worker prefers the firm-preferred outcome). $\mathbf{F} = \{F_1, F_2\}$. $y_{F_1}(N) = 5N$. $y_{F_2}(N) = 3N$. $\mathbf{W} = \{w_1, w_2, w_3\}$. *Amenities are given by this table:*

	F_1	F_2
w_1	5	0
w_2	-1	0
w_3	0	5

We consider two stable outcomes: (μ^1, s^1) and (μ^2, s^2):

$$\mu^1 = \begin{pmatrix} w_1 & w_2 & w_3 \\ F_1 & F_1 & F_2 \end{pmatrix},\ s^1(F_1) = 5,\ s^1(F_2) = 1; \quad \mu^2 = \begin{pmatrix} w_1 & w_2 & w_3 \\ F_1 & F_2 & F_2 \end{pmatrix},\ s^2(F_1) = 0,\ s^2(F_2) = 2.$$

The corresponding profits are

$$\pi_{F_1}\left(\left|\mu^1(F_1)\right|, s^1(F_1)\right) = 2 \times (5-5) = 0, \quad \pi_{F_2}\left(\left|\mu^1(F_2)\right|, s^1(F_2)\right) = 1 \times (3-1) = 2;$$
$$\pi_{F_1}\left(\left|\mu^2(F_1)\right|, s^2(F_1)\right) = 1 \times (5-0) = 5, \quad \pi_{F_2}\left(\left|\mu^2(F_2)\right|, s^2(F_2)\right) = 2 \times (3-2) = 2.$$

Thus, $(\mu^2, s^2) \succeq_{\mathbf{F}} (\mu^1, s^1)$. However, worker w_3 strictly prefers (μ^2, s^2) to (μ^1, s^1).

We now confirm that both outcomes are stable. Both have No Firing and No Envy. The only plausible threat to (μ^1, s^1) having No Poaching would be if F_2 poaches w_2. This would require that F_2 pay $s' \geq 5-1$ which is greater than its marginal product 3. The only plausible threat to (μ^2, s^2) having No Poaching would be if F_1 poaches w_2. This would require that F_1 pay $s' \geq 3$, making profit no greater than 4. This is less than its profit under (μ^2, s^2).

The intuition behind Example C.1 is that moving from one stable outcome to another can make some firms grow while making others shrink, in a manner that all firms benefit. The growing firm increases its salaries to attract marginal workers. This benefits inframarginal workers. The shrinking firms decrease their salaries, which harms those firms' workers.

A final point to emphasize about Example C.1 is its consistency with Proposition 6 and Theorem 2. Neither (μ^1, s^1) nor (μ^2, s^2) is the worker-optimal stable outcome. In the worker-optimal stable outcome (μ^*, s^*): $s^*(F_1) = 5$ and $s^*(F_2) = 3$. Neither firm makes profits at (μ^*, s^*), and all workers are at least as well off as they are in (μ^1, s^1) and (μ^2, s^2). While Theorem 2 implies that workers' preferences are aligned *globally* – there is some outcome which is best for all of them – they need not be aligned *locally*.

C.2 There is no firm-optimal or worker-pessimal stable outcome

The previous subsection demonstrated that payoffs in our model lack the dual lattice structure commonly found in matching models. We will now demonstrate another reason why payoffs in out model lack a dual-lattice structure: there may not be a worker-pessimal or firm-optimal outcome.

Example C.2. $\mathbf{F} = \{F_1, F_2\}$. $\mathbf{W} = \{w_1, w_2, w_3\}$. $y_{F_1}(N) = y_{F_2}(N) = 4N$. *Amenities are given by this table:*

	F_1	F_2
w_1	5	0
w_2	0	5
w_3	0	0

Upon inspection, it is clear that in every stable outcome, all workers will be employed, with $\mu(w_1) = F_1$, and $\mu(w_2) = F_2$.

Let's first consider stable outcomes (μ, s) in which $\mu(F_1) = \{w_1, w_3\}$ and $\mu(F_2) = \{w_2\}$. Having No Poaching requires that firm F_2 be unwilling to pay salary $s(F_1)$ to poach worker w_3:

$$\pi_{F_2}\big(2, s(F_1)\big) < \pi_{F_2}\big(1, s(F_2)\big).$$

Given the firms' production functions, this is equivalent to the requirement that $8 - 2s(F_1) < 4 - s(F_2)$, which in turn is equivalent to the requirement that $4 + s(F_2) < 2s(F_1)$. As such, given that $\mu(F_1) = \{w_1, w_3\}$ and $\mu(F_2) = \{w_2\}$, the minimal value of $s(F_2)$ consistent with having No Poaching is obtained when $s(F_2) = 0, s(F_1) > 2$. Such salaries would also imply the outcome (μ, s) has No Envy and No Firing provided $s(F_1) \leq 4$. These salaries yield firm F_2 profit $\pi_{F_2}(1, 0) = 4$ and yield firm F_1 profit $\pi_{F_1}\big(2, s(F_1)\big) = 8 - 2s(F_1) < 4$.

Now consider stable outcomes (μ', s') with $\mu'(F_1) = \{w_1\}$ and $\mu'(F_2) = \{w_2, w_3\}$. Symmetrically, such outcomes will be stable when $s'(F_2) > 2$, and $s'(F_1) = 0$. This yields firm F_1 profit $\pi_{F_1}(1, 0) = 4$ and yields firm F_2 profit $\pi_{F_2}\big(2, s'(F_2)\big) < 4$. This demonstrates that the stable outcome which is optimal for firm F_1 differs from the stable outcome which is optimal for firm F_2. Thus, while Theorem 2 told us that there is a worker-optimal stable outcome, we see here that there is no firm-optimal stable outcome.

Given that in every stable outcome worker w_1 is matched to firm F_1 and worker w_2 is matched to firm F_2, these workers preferences over stable outcomes depend only on $s(F_1)$ and $s(F_2)$ respectively. In this example, there are stable outcomes in which each of $s(F_1)$ and $s(F_2)$ are equal to 0, but no stable outcome in which they are both equal to 0. Thus, the example also demonstrates that there is no worker-pessimal stable outcome.

C.3 Existing substitutes conditions

Hatfield and Milgrom (2005) present a model which nests both the Gale and Shapley (1962) college admissions model and the Kelso and Crawford (1982) job matching model. They show that a substitutes condition guarantees the existence of a stable matching. In the same model, Hatfield and Kojima (2008) demonstrate a sense in which a weaker substitutes condition is necessary to guarantee the existence of a stable matching. In this subsection, we show that our gross substitutes condition implies neither the Hatfield and Milgrom (2005) substitutes condition nor the Hatfield and Kojima (2008) weak substitutes condition.

The Hatfield and Milgrom model studies contracting between a set of 'hospitals' (i.e., firms) and 'doctors' (i.e., workers). A **contract** $x \in X$ is 'bilateral', and is thus associated with a single doctor x_D and a single hospital x_H. Contracts may be also associated with additional characteristics, such as a salary. Given any hospital h and subset of contracts $X' \subseteq X$, the **chosen set** $C_h(X') \subseteq X'$ represents h's preferred subset of contracts. Hospital h's **rejected set** $R_h(X')$ is the complement of its chosen set: $R_h(X') \equiv X' \setminus C_h(X')$.

Contracts are **substitutes** for hospital h if for all subsets $X' \subseteq X'' \subseteq X$ we have $R_h(X') \subseteq R_h(X'')$. Contracts are **weak substitutes** for hospital h if for those subsets $X' \subseteq X'' \subseteq X$ such that, for all $x, y \in X''$, $x_D = y_D$ implies $x = y$, we have $R_h(X') \subseteq R_h(X'')$.

In other words, contracts are not substitutes if expanding the set of potential contracts means that some contract is no longer rejected. The weak substitutes condition is identical, but it only considers expanded sets of potential contracts containing each doctor at most once.

Our model can be represented in the Hatfield and Milgrom framework as follows. Let a contract x be a hospital-doctor-salary tuple $(x_H, x_D, x_s) \in X = \mathbf{F} \times \mathbf{W} \times \mathbb{R}^+$. A hospital $h \in \mathbf{F}$ selects the chosen set

$$
\begin{aligned}
C_h(X') = \operatorname*{argmax}_{X'' \subseteq X'} \left\{ y_h\left(\left|X''\right|\right) - \sum_{x \in X''} x_s \right\} \text{ subject to } & \forall x \in X'' : x_H = h; \\
& \forall x, x' \in X'' : x \neq x' \implies x_D \neq x'_D; \\
& \forall x, x' \in X'' : x_s = x'_s.
\end{aligned}
$$

The conditions $\forall x \in X'' : x_H = h$ (requiring that a hospital picks only contracts involving itself) and $\forall x, x' \in X'' : x \neq x' \implies x_D \neq x'_D$ (requiring that a hospital picks only one contract involving each doctor) are imposed by Hatfield and Milgrom. Our additional requirement $\forall x, x' \in X'' : x_s = x'_s$ requires that hospitals set homogeneous salaries.

Our gross substitutes condition does not guarantee that these chosen sets will satisfy the Hatfield and Milgrom substitutes condition. For example, consider a hospital h with constant marginal product $y_h(N) = 4N$. Let there be three doctors d_1, d_2, d_3. Consider the set of contracts

$$
X'' = \left\{(h, d_1, 1), (h, d_2, 2), (h, d_3, 2)\right\}.
$$

X'' represents a context in which the hospital can pay salary 1 to hire doctor d_1 or can pay salary 2 to hire either doctor d_2 or d_3. Hiring only doctor d_1 at salary 1 yields the hospital a profit of $4 - 1 = 3$ whereas hiring both d_2 and d_3 at salary 2 yields the hospital a profit of $8 - 4 = 4$. Thus,

$C_h(X'') = \{(h, d_2, 2), (h, d_3, 2)\}$ and $R_h(X'') = \{(h, d_1, 1)\}$. However, if the hospital can only hire either d_1 or d_2, as represented by

$$X' = \{(h, d_1, 1), (h, d_2, 2)\},$$

then it will prefer to do so at the minimal possible salary: $C_h(X') = \{(h, d_1, 1)\}$ and $R_h(X') = \{(h, d_2, 2)\}$. Thus, $X' \subseteq X''$ but $R_h(X') \nsubseteq R_h(X'')$, breaching both the substitutes condition and the weak substitutes condition.

That our gross substitutes condition does not imply the Hatfield and Milgrom substitutes condition means that our results do not follow trivially from existing work.

That our gross substitutes condition does not imply the Hatfield and Kojima weak substitutes condition may be surprising: Hatfield and Kojima show that, when some hospital's preferences fail the weak substitutes condition, one can construct an example such that a stable outcome does not exist. This might seem to contradict our Theorem 1, which implied that a stable outcome always exists. The supposed contradition is resolved by noting that Hatfield and Kojima's result requires the existence of a second hospital with strict preferences over different doctors. Such a hospital is ruled out by our model, which assumes that hospitals view doctors as interchangeable.

C.4 The core

An outcome is in the core if no coalition of workers and firms can deviate from it, producing more value than the sum of their payoffs in the original outcome. In many matching models, the core coincides with stability, leading to the two terms to be used somewhat interchangeably. For example, Kelso and Crawford refer to their solution concept as the core. Our definition of stability requires that, in a blocking coalition, every worker receives the same salary. This restriction on transfers means that the core does not coincide with stability. One implication is that stable outcomes can be inefficient (Example 1). A core outcome can never be inefficient: if it were, the coalition consisting of every worker and firm could deviate from it.

C.5 Pairwise stability

Pairwise stability requires that no worker-firm pair can unilaterally deviate such that both are better off (Roth & Sotomayor, 1990). In some many-to-one matching models, an outcome (or, in models without salaries, simply a matching) is stable if and only if it is pairwise stable. That equivalence between pairwise stability and stability does not hold in our model.

Proposition 1 told us that stable outcomes must satisfy No Poaching: no firm can unilaterally increase its salary, attract more workers, and make at least as much profit. Higher salaries must be paid to a firm's existing workers as well as to the workers that it poaches. A firm may be willing to increase its salary when doing so would attract many workers, but not when doing so would attract only a single worker. Thus, pairwise stability is a weaker requirement than stability.

C.6 Competitive equilibria

An outcome (μ, s) is a **competitive equilibrium** if

$$\forall F \in \mathbf{F} : \left|\mu(F)\right| \in \underset{L \in \mathbb{N}}{\arg\max} \left\{\pi_F\big(L, s(F)\big)\right\}, \tag{34}$$

$$\forall w \in \mathbf{W} : \mu(w) \in \underset{F \in \mathbf{F} \cup \{\emptyset\}}{\arg\max} \left\{\alpha_w(F) + s(F)\right\}. \tag{35}$$

In a competitive equilibrium, firms choose quantities taking salaries as fixed, while workers choose firms taking salaries as fixed.

Lemma C.1. *An outcome is a competitive equilibrium if and only if it has both Marginal Product Salaries and No Envy.*

Proof. Expression (35) is equivalent to requiring that the outcome has No Envy. It thus remains for us to show that expression (34) is equivalent to requiring that the outcome has Marginal Product Salaries.

We first show that an outcome satisfying equation (34) has Marginal Product Salaries. Let (μ, s) be an outcome. If for some firm F: $s(F) > \Delta^-_\mu(F) = y_F\left(\left|\mu(F)\right|\right) - y_F\left(\left|\mu(F)\right| - 1\right)$ then $y_F\left(\left|\mu(F)\right|\right) - y_F\left(\left|\mu(F)\right| - 1\right) - s(F) < 0$. Thus

$$\begin{aligned}
\pi_F\left(\left|\mu(F)\right|, s(F)\right) &= y_F\left(\left|\mu(F)\right|\right) - s(F)\left|\mu(F)\right| \\
&= y_F\left(\left|\mu(F)\right| - 1\right) - s(F)\left(\left|\mu(F)\right| - 1\right) + y_F\left(\left|\mu(F)\right|\right) - y_F\left(\left|\mu(F)\right| - 1\right) - s(F) \\
&< y_F\left(\left|\mu(F)\right| - 1\right) - s(F)\left(\left|\mu(F)\right| - 1\right) \\
&= \pi_F\left(\left|\mu(F)\right| - 1, s(F)\right),
\end{aligned}$$

and thus equation (34) fails to hold. A similar contradiction arises if $s(F) < \Delta^+_\mu(F) = y_F\left(\left|\mu(F)\right| + 1\right) - y_F\left(\left|\mu(F)\right|\right)$. By the contrapositive, an outcome satisfying equation (34) has Marginal Product Salaries.

We next show that if an outcome has Marginal Product Salaries, then it satisfies equation (34). Let equation (34) fail for (μ, s): there exists L' such that

$$\pi_F\left(L', s(F)\right) > \pi_F\left(\left|\mu(F)\right|, s(F)\right). \tag{36}$$

Consider first the case where $L' > \left|\mu(F)\right|$. Inequality (36) implies that $y_F(L') - y_F\left(\left|\mu(F)\right|\right) > s(F)\left(L' - \left|\mu(F)\right|\right)$. By Lemma 1, y_F has decreasing differences, which implies that $y_F(L') - y_F\left(\left|\mu(F)\right|\right) \leq$

$\Delta^+_\mu(F)\left(L' - \left|\mu(F)\right|\right)$. Together, these inequalities imply that

$$\Delta^+_\mu(F)\left(L' - \left|\mu(F)\right|\right) > s(F)\left(L' - \left|\mu(F)\right|\right),$$

which with $L' > \left|\mu(F)\right|$ implies that $\Delta^+_\mu(F) > s(F)$. The outcome thus lacks Marginal Product Salaries.

Next consider the case where $L' < \left|\mu(F)\right|$. Inequality (36) implies that $y_F\left(\left|\mu(F)\right|\right) - y_F(L') < s(F)\left(\left|\mu(F)\right| - L'\right)$. By Lemma 1, y_F has decreasing differences, which implies that $\Delta^-_\mu(F)\left(\left|\mu(F)\right| - L'\right) \leq y_F\left(\left|\mu(F)\right|\right) - y_F(L')$. Together, these inequalities imply that

$$\Delta^-_\mu(F)\left(\left|\mu(F)\right| - L'\right) < s(F)\left(\left|\mu(F)\right| - L'\right),$$

which with $L' < \left|\mu(F)\right|$ implies that $\Delta^-_\mu(F) < s(F)$. This is again is inconsistent with the outcome having Marginal Product Salaries. Thus, an outcome failing equation (34) must lack Marginal Product Salaries. By the contrapositive, if an outcome has Marginal Product Salaries, then it must also satisfy equation (34). □

Together with Lemma 6, Lemma C.1 implies that the worker-optimal stable outcome is a competitive equilibrium. As such, a competitive equilibrium exists. Lemma C.1 also allows us to characterize the set of competitive equilibria: by Proposition 2 they are stable and efficient. In sum, we have the following corollary:

Corollary C.1. *A competitive equilibrium exists. Moreover, if (μ, s) is a competitive equilibrium, then (μ, s) is a stable outcome, and μ is efficient.*

Competitive equilibria treat firms as naïve. Given an efficient outcome, a firm does not realize that hiring inefficiently few workers would let it pay lower salaries. Given an inefficient outcome, a firm will demand more workers than are willing to work at the prevailing salary. Thus, competitive equilibria are necessarily efficient.

In a stable outcome, firms cannot unilaterally reduce salaries: doing so would require the consent of their existing workers. In this sense, stability also forces firms to take the salary level as given (although a firm can always increase its salary). Theorem 1 told us that this mechanism prevents firms from destabilizing an efficient outcome. However, stability does let firms understand that they cannot employ arbitrarily many workers at the prevailing salary. Firms can thus resist the temptation to destabilize an inefficient outcome: while an efficient outcome is stable, inefficient outcomes may be as well. This suggests that inefficient stable outcomes are caused by firms failing to take salaries as given.

Although Corollary C.1 connects competitive equilibria and efficient stable outcomes, they are not equivalent. Every competitive equilibrium is an efficient stable outcome, but not every efficient stable outcome is a competitive equilibrium. For example, a monopsonist might be able to employ all available workers at a salary below their marginal product. This could be an efficient stable outcome but could never be a competitive equilibrium because, at that salary, the monopsonist would prefer to employ additional workers. Thus, stable outcomes can yield efficient quantities without getting prices 'right'.

Similarly, it's worth noting that there could be many competitive equilibria. Because they are efficient, they will generically have the same matching of workers to firms, but they can have differing salary schedules. This is a limitation of Kojima (2007)'s analysis. Kojima argues that strategic salary-setting by firms can be better for inframarginal workers than a competitive equilibrium, as firms increase salaries to compete for marginal workers. Kojima limits his comparisons to the firm-optimal competitive equilibrium. This perspective is limiting: no worker benefits from firms' strategic salary-setting when it differs from the worker-optimal stable outcome. By Corollary C.1, the worker-optimal stable outcome is also a competitive equilibrium.

C.7 Bertrand equilibria

The Bertrand salary-setting game is a two-stage game. In the first period, firms simultaneously choose salaries. In the second period, each worker chooses a firm. Thus, each firm F's strategy is $s_F \in \mathbb{R}$ while each worker w's strategy is a function, which takes as input the vector of salaries and selects a firm:

$$\mathrm{Ch}_w : \mathbb{R}^{|\mathbf{F}|} \to \mathbf{F} \cup \{\emptyset\}.$$

Let $s \equiv (s_F)_{F\in\mathbf{F}}$ denote the vector of salaries chosen by firms. Let $\mathrm{Ch} \equiv (\mathrm{Ch}_w)_{w\in\mathbf{W}}$ denote the vector of choice functions chosen by workers. Let $L_F^\circ(s_F, s_{-F}, \mathrm{Ch})$ denote the number of workers for whom $\mathrm{Ch}_w = F$, given the vector of salaries with firm F's element equal to s_F and other elements equal to the corresponding element of $s_{-F} \equiv (s_{F'})_{F'\neq F}$. Note that $L_F^\circ(\cdot)$ differs from our definition of $L_F(\cdot)$ in Section 2.4: $L_F(\cdot)$ allocated a worker indifferent between two firms to both, whereas $L_F^\circ(\cdot)$ allocates such a worker to only one.

A **Bertrand equilibrium** (Ch^*, s^*) is a subgame perfect pure strategy Nash equilibrium of the Bertrand salary-setting game. It comprises a vector of choice functions Ch^* and a vector of salaries s^*. Firms set salaries optimally, given the other firms' salaries and the workers' choice functions:

$$\forall F : s_F^* \in \underset{s_F\in\mathbb{R}^+}{\arg\max} \left\{\pi_F\left(L_F^\circ\left(s_F, s_{-F}^*, \mathrm{Ch}^*\right), s_F\right)\right\}. \tag{37}$$

Workers' choice functions are optimal given all possible salaries:

$$\forall w : \forall s : \mathrm{Ch}_w^*(s) \in \underset{F\in\mathbf{F}\cup\emptyset}{\arg\max} \left\{\alpha_w(F) + s_F\right\}. \tag{38}$$

To connect this solution concept to our earlier analysis, we say that an outcome (μ, s) is a Bertrand equilibrium if there exists a Bertrand equilibrium (Ch, s) such that $\forall w : \mathrm{Ch}_w(s) = \mu(w)$.

We will focus on the case where all firms have constant returns to scale. The below lemma shows that this yields a simple characterization of Bertrand equilibria. We will use that characterization to show that Bertrand equilibria are stable.

Lemma C.2. *Let all firms have constant returns to scale $y_F(N) = \Delta_F N$. Let (μ, s) be a Bertrand equilibrium. For every firm F: either $\mu(F) = \emptyset$, $s(F) = 0$, or there exists a firm $F' \neq F$ and a worker $w \in \mu(F)$ such that $s(F) = \alpha_w(F') - \alpha_w(F) + \Delta_{F'}$.*

Proof. Consider a firm F with $\mu(F) \neq \emptyset$ and $s(F) > 0$. (Recall salaries are non-negative by definition.) For such a firm to be worse off unilaterally decreasing its salary, such a decrease must cause it to lose a worker. Thus, there must be some worker w and some firm $F' \neq F$ such that $\mathrm{Ch}_w(s(F), s_{-F}) = F$ but, for all $r < s(F)$: $\mathrm{Ch}_w(r, s_{-F}) = F'$. By expression (38) $\alpha_F + s(F) = \alpha_{F'} + s(F')$. The firm F' would benefit by slightly increasing its salary and poaching worker w unless its salary equals its marginal product $\Delta_{F'}$. Thus, $s(F') = \Delta_{F'}$. Combining these two expressions implies that $s(F) = \alpha_w(F') - \alpha_w(F) + \Delta_{F'}$. □

Lemma C.2 highlights the effects of competition in the Bertrand game. No firm will pay a salary such that its workers strictly prefer that firm over other firms: if it did so, it could profitably decrease its salary. Thus, at each firm there will be some worker who is indifferent between working at that firm and working at another firm. That other firm could poach the worker by paying an infinitesimally higher salary; for that to be unprofitable, its salary must already equal its marginal product.

We will use Lemma C.2 to argue that Bertrand equilibria can be more efficient than some stable outcomes. Before doing so, we use it to prove the following proposition, which argues that Bertrand equilibria are themselves generically stable outcomes.

Proposition C.1. *Let each firm F have constant returns to scale $y_F(N) = \Delta_F N$. For almost all technologies Δ_F and amenities $\alpha_w(F)$, all Bertrand equilibria are stable outcomes.*

Proof. Consider a Bertrand equilibrium (μ, s). We will show that (μ, s) has No Envy and No Firing. We will then use Lemma C.2 to show that (μ, s) will only lack No Poaching in a knife-edge case. By Proposition 1, this implies Proposition C.1.

Step 1: (μ, s) has No Envy.

Step 1 follows immediately from expression (38).

Step 2: (μ, s) has No Firing.

Proof of Step 2: Given constant returns to scale, $s(F) > \Delta^-_\mu(F)$ implies that firm F makes negative profits. F would be better off choosing salary $s(F) = 0$ and making non-negative profits. Thus, in every Bertrand equilibrium, $s(F) \leq \Delta^-_\mu(F)$.

Step 3: There is no firm $F \in \mathbf{F}$, salary $s_F > s(F)$ and employment level L such that $\left|\mu(F)\right| < L \leq L_F(s_F, s)$, and that $\pi_F\left(L, s'\right) > \pi_F\left(\left|\mu(F)\right|, s(F)\right)$.

Proof of Step 3: Assume towards a contradiction that there is a firm $F \in \mathbf{F}$, salary $s_F > s(F)$ and employment level L such that $\left|\mu(F)\right| < L \leq L_F(s_F, s)$, and that $\pi_F\left(L, s'\right) > \pi_F\left(\left|\mu(F)\right|, s(F)\right)$. By the assumption that production functions have constant returns to scale:

$$(\Delta_F - s_F)L > \left(\Delta_F - s(F)\right)\left|\mu(F)\right|.$$

Given that $\left|\mu(F)\right| < L \leq L_F(s_F, s)$, this implies that

$$(\Delta_F - s_F)L_F(s_F, s) > \left(\Delta_F - s(F)\right)\left|\mu(F)\right|.$$

Given that this inequality is strict, there exists an $s'_F > s_F$ such that

$$(\Delta_F - s'_F)L_F(s_F, s) > \left(\Delta_F - s(F)\right)\left|\mu(F)\right|.$$

If a worker is indifferent between working at F and some other firm F' at salary s_F, she will strictly prefer working at F at salary s'_F. By expression (38), $L^\circ_F(s'_F, s) \geq L_F(s_F, s)$ and so

$$(\Delta_F - s'_F)L^\circ_F(s'_F, s) > \left(\Delta_F - s(F)\right)\left|\mu(F)\right|,$$

implying that $\left(\mu, s\right)$ is not a Bertrand equilibrium.

Step 4: $\left(\mu, s\right)$ will almost always have No Poaching.

Proof of Step 4: Given Step 3, if $\left(\mu, s\right)$ lacks No Poaching, it will only do so by equality:

$$\exists F_1 \in \mathbf{F}, s_{F_1} > s(F_1), L \in \mathbb{N} \text{ with } \left|\mu(F)\right| < L \leq L_{F_1}(s_{F_1}, s) \text{ such that } \pi_{F_1}(L, s_{F_1}) = \pi_{F_1}\left(\left|\mu(F_1)\right|, s(F_1)\right). \tag{39}$$

Note that if there was some $s'_{F_1} \in (s(F), s_{F_1})$ such that $L_{F_1}(s'_{F_1}, s) = L_{F_1}(s_{F_1}, s)$, Step 3 would fail. Thus, at least one worker $w_1 \notin \mu(F_1)$ must be indifferent between working for some firm $F_2 \in \mathbf{F} \cup \{\emptyset\} \setminus \{F_1\}$ at salary $s(F_2)$ and for firm F_1 at salary s_{F_1}:

$$s_{F_1} = s(F_2) + \alpha_{w_1}(F_2) - \alpha_{w_1}(F_1). \tag{40}$$

If $\mu(F_1) = \emptyset$, then $s(F_1) = y_F(1_1) = \Delta_{F_1}$ and so firm F_1 could never raise its salary, hire workers and make non-negative profit. Thus, $\mu(F_1) \neq \emptyset$. Given Lemma C.2, this tells us that:

$$s(F_1) = 0 \text{ or } \exists F_3 \in \mathbf{F} \cup \{\emptyset\} \setminus \{F_1\}, w_2 \in \mu(F_1) \text{ such that } s(F_1) = \alpha_{w_2}(F_3) - \alpha_{w_2}(F_1) + \Delta_{F_3} \tag{41}$$

$$\text{and } s(F_2) = 0 \text{ or } \exists F_4 \in \mathbf{F} \cup \{\emptyset\} \setminus \{F_2\}, w_3 \in \mu(F_2) \text{ such that } s(F_2) = \alpha_{w_3}(F_4) - \alpha_{w_3}(F_2) + \Delta_{F_4}. \tag{42}$$

Combining equations (39)-(42), the following must hold:

$$
\begin{aligned}
&\left(L-\left|\mu(F_1)\right|\right)\Delta_{F_1} - L\left(\alpha_{w_1}(F_2)-\alpha_{w_1}(F_1)\right)\\
&\qquad = \begin{cases}
0 & \text{if } s(F_1)=s(F_2)=0;\\
-\left|\mu(F_1)\right|\left(\alpha_{w_2}(F_3)-\alpha_{w_2}(F_1)+\Delta_{F_3}\right) & \text{if } s(F_1)\neq 0, s(F_2)=0;\\
L\left(\alpha_{w_3}(F_4)-\alpha_{w_3}(F_2)+\Delta_{F_4}\right) & \text{if } s(F_1)=0, s(F_2)\neq 0;\\
L\left(\alpha_{w_3}(F_4)-\alpha_{w_3}(F_2)+\Delta_{F_4}\right)-\left|\mu(F_1)\right|\left(\alpha_{w_2}(F_3)-\alpha_{w_2}(F_1)+\Delta_{F_3}\right) & \text{if } s(F_1)\neq 0, s(F_2)\neq 0.
\end{cases}
\end{aligned}
$$

That condition is, admittedly, quite opaque. For our purposes, its critical property is that is expressed solely in terms of technologies, amenities and the integer-valued L and $\left|\mu(F_1)\right|$. Thus, the amenities and technologies for which such an expression can hold have measure 0. This implies that for almost all amenities and technologies, if the Bertrand equilibrium lacks No Poaching, then it lacks it by strict inequality. We showed above that if the an outcome lacks No Poaching by strict inequality, then that outcome is not a Bertrand equilibrium. □

The critical distinction between a Bertrand equilibrium and other stable outcomes is that, in a Bertrand equilibrium, firms can unilaterally decrease their salaries. Interestingly, this does not imply that Bertrand equilibria are less efficient or better for firms than other stable outcomes.

This point can be demonstrated by revisiting Example C.2. Let (μ, s) be a Bertrand equilibrium. No firm will pay strictly more than the other: if $s(F_1) > s(F_2)$, for example, firm F_1 would retain its current workers by paying any salary $s' \in \left(s(F_2), s(F_1)\right)$. Such a salary would increase firm F_1's profit. Thus, $s(F_1) = s(F_2)$, and so worker w_3 will be indifferent between the two firms. If $s(F_1) < 4$ and $\mu(w_3) = F_2$, firm F_1 could pay an infinitesimally higher salary $s' \in \left(s(F_1), 4\right)$, employ w_3 and make profit $2 \times (4 - s') > 4 - s(F_1)$. Similarly, if $s(F_2) < 4$ and $\mu(w_3) = F_1$ then firm F_2 could profit by paying an infinitesimally higher salary. Thus, in every Bertrand equilibrium, $s(F_1) = s(F_2) = 4$, and both firms make zero profit.

Consider this other outcome:

$$\mu' = \begin{pmatrix} w_1 & w_2 & w_3 \\ F_1 & F_2 & F_2 \end{pmatrix},\ s'(F_1) = 0,\ s'(F_2) = 3.$$

At outcome $\left(\mu', s'\right)$, both firms make positive profit: $\pi_{F_1} = 4$; $\pi_{F_2} = 2$. Yet $\left(\mu', s'\right)$ is a stable outcome: that this outcome has No Envy and No Firing is self-evident, while it has No Poaching because firm F_1 would have to pay salary 3 to poach w_3, which would yield F_1 a profit of 2.

In Bertrand competition, firms are constantly tempted to decrease their salaries. This can destabilize profitable outcomes, eventually making all firms worse off. In a stable outcome, firms cannot decrease their salaries in a blocking coalition while retaining their existing workers. Example C.2 shows that firms can benefit from an inability to decrease their salaries.

However, Bertrand equilibria are not necessarily efficient or worker-optimal. In Example 1, the unique Bertrand equilibrium has the firm paying salary zero. The efficiency of Bertrand equilibria depends on whether firms can be induced to compete for marginal workers.

D No Stable Outcome with Non-Interchangeable Workers

Throughout the main text we assumed that workers are interchangeable in production: a firm's output depended only on the number of workers it employed. In this appendix, we show that a stable outcome may not exist when that assumption is dropped. In fact, a stable outcome may not exist in the simple case where firms have homogeneous technologies which are additive in workers' productivities, and where there are no worker-firm amenities. We present an example in which a stable outcome does not exist. To simplify our exposition — and because the case may be of independent interest — we first characterize stable outcomes in this simple case.

This model comprises a set of firms **F** and a set of workers **W**. There are fewer firms than workers. Each worker $w \in \mathbf{W}$ is endowed with a productivity $\rho_w > 0$. Firms are symmetric and have output equal to the sum of the productivities of the workers to which they are matched. Thus, if firm F employs workers $C \subseteq W$ at salary s its profit will be

$$\pi_F(C, s) = \sum_{w \in C} \left[\rho_w - s\right].$$

Workers care only about their salary: $u_w(F, s) = s$. We consider the generic case where each worker's productivity is different to that of every other: $w \neq w' \implies \rho_w \neq \rho_{w'}$.

Matchings and outcomes are defined as in the main text, and the definition of a stable outcome is unchanged.

Proposition D.1. *Any stable outcome can be characterized by a labeling of the M firms $1, 2, \ldots, M$ and a set of intervals $\{[0, s_1), [s_1, s_2), \ldots, [s_M, s_{M+1})\}$, where $s_j < s_{j+1}$ and $s_{M+1} = \infty$. The firm labeled j will pay salary s_j and will hire all workers with productivity in $[s_j, s_{j+1})$. Workers with productivity in $[0, s_1)$ will be unemployed. Moreover, all firms must make the same profit and firms must be making profit no less than the sum of unemployed worker productivities.*

Proof: Let (μ, s) denote a stable outcome. We prove Proposition D.1 in seven steps.

Step 1: $\forall F : \forall w \in \mu(F) : \rho_w \geq s(F)$.

Proof of Step 1: Otherwise, $(F, \mu(F) \setminus \{w\}, s(F))$ would block (μ, s).

Step 2: If $s(F) < s(F')$ and $\mu(w) = F$, $s(F') > \rho_w$.

Proof of Step 2: Otherwise, $(F', \mu(F') \cup \{w\}, s(F'))$ would block (μ, s).

Step 3: $\forall F : \mu(F) \neq \emptyset$.

Proof of Step 3: By Step 1, $\forall w : \rho_w \geq s(\mu(w))$. Moreover, each ρ_w differs and there are fewer firms than workers. Thus, there is some worker w for whom $\rho_w > s\big(\mu(w)\big)$. If $\exists F : \mu(F) = \emptyset$, $\big(F, \{w\}, \rho_w\big)$ would block $\big(\mu, s\big)$.

Step 4: $\forall F \neq F' : s(F) \neq s(F')$.

Proof of Step 4: Assume $s(F) = s(F')$. By Step 3, both firms are matched to at least one worker. There is at most one worker with $\rho_w = s(F)$. Thus, by Step 1, at least one of F or F' must be matched to a worker w' with $\rho_{w'} > s(F) = s(F')$. If $\mu(w') = F$ then $\big(F', \mu(F') \cup \{w'\}, s(F')\big)$ would block $\big(\mu, s\big)$. If $\mu(w') = F'$ then $\big(F, \mu(F) \cup \{w'\}, s(F)\big)$ would block $\big(\mu, s\big)$.

Step 5: If $\mu(w) = \emptyset : \forall F : \rho_w < s(F)$.

Proof of Step 5: Otherwise, $\big(F, \mu(F) \cup \{w\}, s(F)\big)$ would block $\big(\mu, s\big)$.

Step 6: All firms must make the same profit.

Proof of Step 6: If firm F made more profit than firm F', then $\big(F', \mu(F), s(F)\big)$ would block $\big(\mu, s\big)$.

Step 7: Each firm's profit must be no less than the sum of unemployed worker productivities.

Proof of Step 7: Let C denote the set of unemployed workers. If $\sum_{w \in C} \rho_w > \pi_F\big(\mu(F), s(F)\big)$, then $(F, C, 0)$ would block $\big(\mu, s\big)$.

Step 4 implies that firm salaries form disjoint intervals $[s_1, s_2), [s_2, s_3), \ldots$. Steps 1, 2 and 5 imply that the firm paying salary s_j will hire all workers with productivity in $[s_j, s_{j+1})$. Steps 6 and 7 correspond to Proposition D.1's final sentence. □

Given Proposition D.1, it is relatively simple to produce an example without a stable outcome. The following is such an example.

Example D.1 (with non-interchangeable workers, a stable outcome need not exist). $\mathbf{F} = \{F_1, F_2, F_3\}$. $\mathbf{W} = \{1, 2, 5, 6\}$. *Each worker is labeled by their productivity:* $\forall w \in \mathbf{W} : \rho_w = w$.

We will now show that Example D.1 lacks a stable outcome. We will consider each candidate matching in turn, exploiting Proposition D.1 to limit the number of matchings we must consider. As firms are homogeneous it is without loss of generality to assume that firm F_1 employs the least productive employed worker(s) and that firm F_3 employs the most productive.

Candidate matching 1: $\begin{pmatrix} 1 & 2 & 5 & 6 \\ \emptyset & F_1 & F_2 & F_3 \end{pmatrix}$.

$\mu(1) = \emptyset$, and thus, by Proposition D.1, each firm must make profit no less than 1. However, it is impossible for firm F_3 to make profit no less than 1, as Proposition D.1 implies that $s(F_3) > 5$.

Candidate matching 2: $\begin{pmatrix} 1 & 2 & 5 & 6 \\ F_1 & F_1 & F_2 & F_3 \end{pmatrix}$.

By Proposition D.1, $s(F_1) \leq 1$, and thus $\pi_{F_1} \geq 1$. However, it is again impossible for firm F_3 to make profit no less than 1, as Proposition D.1 implies that $s(F_3) > 5$.

Candidate matching 3: $\begin{pmatrix} 1 & 2 & 5 & 6 \\ F_1 & F_2 & F_2 & F_3 \end{pmatrix}$.

By Proposition D.1, $s(F_2) \leq 2$, and thus $\pi_{F_2} \geq 3$. However, it is impossible for firm F_3 to make profit no less than 3, as Proposition D.1 implies that $s(F_3) > 5$.

Candidate matching 4: $\begin{pmatrix} 1 & 2 & 5 & 6 \\ F_1 & F_2 & F_3 & F_3 \end{pmatrix}$.

By Proposition D.1, $s(F_3) \le 5$, and thus $\pi_{F_3} \ge 1$. However, it is impossible for firm F_2 to make profit no less than 1, as Proposition D.1 implies that $s(F_2) > 1$. This argument completes the proof that Example D.1 lacks a stable outcome.

If the productivities in this example are perturbed, a stable outcome exists. This observation suggests that a stable outcome may almost-always exist. It also suggests that there might always exist a weak stable outcome, which is defined to consider only a breaking coalition of workers and firms who are *all* strictly better off compared to the candidate outcome. We leave investigation of these conjectures for future work.

E Duplicating Workers

Proposition 5 tells us that when each firm has a duplicate, every stable outcome is efficient. In this appendix, we show that the effect of duplicating workers is very different: in some sense, nothing happens when workers are duplicated. Of course, given decreasing returns to scale, increasing the number of workers may decrease firms' marginal products and thus decrease salaries. We focus on the effect of workers' 'market power' by duplicating workers while 'stretching' firms' production function appropriately. This duplication and stretching has no effect on the set of stable salary schedules.

Before formalizing that result we need to formalize the relationship between two labor markets. Consider two labor markets $(\mathbf{F}, \mathbf{W})$ and $(\mathbf{G}, \mathbf{X})$. Let $\mathscr{P}(\mathbf{X})$ denote the power set of $\mathbf{X}$. We say that $\psi : \mathbf{W} \cup \mathbf{F} \to \mathscr{P}(\mathbf{X} \cup \mathbf{G})$ is a **transformation** from $(\mathbf{F}, \mathbf{W})$ to $(\mathbf{G}, \mathbf{X})$ if $\{\psi(w) : w \in \mathbf{W}\}$ partitions $\mathbf{X}$ while $\{\psi(F) : F \in \mathbf{F}\}$ partitions $\mathbf{G}$.

We are interested in comparing labor markets in which all workers are duplicated and all firms are stretched. A transformation ψ from $(\mathbf{F}, \mathbf{W})$ to $(\mathbf{G}, \mathbf{X})$ *duplicates workers and stretches firms* if

$$\forall F \in \mathbf{F} : \left|\psi(F)\right| = 1;$$

$$\forall w \in \mathbf{W} : \left|\psi(w)\right| = 2;$$

$$\forall w \in \mathbf{W}, x \in \psi(w), F \in \mathbf{F} : \alpha_w(F) = \alpha_x\left(\psi(F)\right);$$

$$\forall F \in \mathbf{F}, N \in \mathbb{N} : y_{\psi(F)}(N) = \sum_{i=1}^{N} \left[y_F\left(\lceil i \div 2 \rceil\right) - y_F\left(\lceil i \div 2 \rceil - 1\right) \right], \text{ where } \lceil \cdot \rceil \text{ is the ceiling function.}$$

The first condition requires that there be one firm in $\mathbf{G}$ for every firm in $\mathbf{F}$. The second condition requires that there be two workers in $\mathbf{X}$ for every worker in $\mathbf{W}$. The third condition requires that firms in **G** provide workers the same amenities as the corresponding firms in **F**. The fourth condition requires that firms in **G** have production functions similar to those in **F** but stretched so that each marginal product can be produced by each of two workers.

Proposition E.1. *Let ψ be a transformation from $(\mathbf{F},\mathbf{W})$ to $(\mathbf{G},\mathbf{X})$ that duplicates workers and stretches firms. Let (μ, s) be an outcome in the labor market $(\mathbf{F},\mathbf{W})$, and let (μ', s') be an outcome in the labor market $(\mathbf{G},\mathbf{X})$ such that*

$$\forall F \in \mathbf{F} : s(F) = s'\left(\psi(F)\right);\ \textit{and}\ \forall w \in \mathbf{W}, x \in \psi(w) : \mu'(x) = \psi\left(\mu(w)\right).$$

(μ, s) is a stable outcome if and only if (μ', s') is a stable outcome.

Proof. We will show that the No Envy, No Firing, and No Poaching conditions are equivalent across the two outcomes. By Proposition 1, this equivalence implies Proposition E.1.

Step 1: (μ, s) has No Envy if and only if (μ', s') has No Envy.
Proof of Step 1: Consider workers $w \in \mathbf{W}, x \in \psi(w)$ and firms $F, F' \in \mathbf{F}, G = \psi(F), G' = \psi(F')$. Given that $s(F) = s'(G)$, $s(F') = s'(G')$, $\alpha_w(F) = \alpha_x(G)$ and $\alpha_w(F') = \alpha_x(G')$:

$$\alpha_w(F) + s(F) \geq \alpha_w(F') + s(F') \iff \alpha_x(G) + s'(G) \geq \alpha_x(G') + s'(G').$$

Thus, the No Envy conditions for the two outcomes are equivalent.

Step 2: (μ, s) has No Firing if and only if (μ', s') has No Firing.
Proof of Step 2: Letting $G = \psi(F)$:

$$\begin{aligned}\Delta^-_{\mu'}(G) &= y_F\left(\left\lceil\left|\mu'(G)\right| \div 2\right\rceil\right) - y_F\left(\left\lceil\left|\mu'(G)\right| \div 2\right\rceil - 1\right) \\ &= y_F\left(\left|\mu'(F)\right|\right) - y_F\left(\left|\mu'(F)\right| - 1\right) \\ &= \Delta^-_{\mu}(F),\end{aligned}$$

where the second equality follows from two workers being matched to G for every one matched to F, and thus $\left\lceil\left|\mu'(G)\right| \div 2\right\rceil = \left\lceil 2 \times \left|\mu'(F)\right| \div 2\right\rceil = \left|\mu'(F)\right|$. Given that $s(F) = s'(G)$, it follows that

$$s(F) \leq \Delta^-_{\mu}(F) \iff s'(G) \leq \Delta^-_{\mu'}(G).$$

Step 3: (μ, s) has No Poaching if and only if (μ', s') has No Poaching.
Proof of Step 3: For some firm $F \in \mathbf{F}$, let $G = \psi(F)$. Note that for any $L \in \mathbb{N}$:

$$\begin{aligned}y_G(2L) &= \sum_{i=1}^{2L} \left[y_F\left(\lceil i \div 2\rceil\right) - y_F\left(\lceil i \div 2\rceil - 1\right)\right] \\ &= \sum_{j=1}^{L} 2\left[y_F\left(\lceil j\rceil\right) - y_F\left(\lceil j\rceil - 1\right)\right] = 2y_F(L).\end{aligned}$$

It follows that for any salary r:

$$\pi_G(2L, r) = y_G(2L) - r \times 2L = 2\pi_F(L, r).$$

In particular, given that $\left|\mu'(G)\right| = 2\left|\mu(F)\right|$ and $s(F) = s'(G)$: $\pi_G\left(\left|\mu'(G)\right|, s'(G)\right) = 2\pi_F\left(\left|\mu(F)\right|, s(F)\right)$.

Let (μ, s) lack No Poaching because of firm F: there exists $r > s(F)$ and $L \leq L_F(r, s)$ with $L > \left|\mu(F)\right|$ such that $\pi_F(L, r) \geq \pi_F\left(\left|\mu(F)\right|, s(F)\right)$. By the above, $\pi_G(2L, r) = 2\pi_F(L, r)$ and $\pi_G\left(\left|\mu'(G)\right|, s'(G)\right) = 2\pi_F\left(\left|\mu'(F)\right|, s(F)\right)$. Thus:

$$\pi_G(2L, r) \geq \pi_G\left(\left|\mu'(G)\right|, s'(G)\right).$$

Moreover, $L_G(r, s') = 2L_F(r, s)$. Thus, $2L \leq L_G(r, s')$. Thus, (μ', s') lacks No Poaching. By the contrapositive, if (μ', s') has No Poaching, then (μ, s) has No Poaching. The proof of the converse is symmetric. □

If firms have constant returns to scale – as they do in all of our examples – 'stretching' firms does not change them. This observation motivates a simpler version of Proposition E.1: Assume that each firm has constant returns to scale. If each worker is duplicated while each firm is unchanged, the set of stable outcomes will be unchanged, except that the two duplicate workers take the place of the one original worker. Thus, every example in this paper can be extended to involve arbitrarily many workers, with the nature of the example unchanged.